Pairwise contraction characterization of orthogonally decomposable symmetric tensors

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Editorial note. Only thm:odeco counts. Full author text and proofs retained; orthogonal Holant converse is prerequisite and not counted again.

Editorial note. Rising diagonal dots implemented using reflected standard ddots; BibLaTeX-comment-only lines omitted from derived BibTeX bibliography, original retained.

Allowed standard prerequisites (editorial).

- Standard graph algorithms and ETH cited by authors. All source-local proofs and necessary context retained.

Abstract

The Holant theorem is a powerful tool for studying the computational complexity of counting problems. Due to the great expressiveness of the Holant framework, a converse to the Holant theorem would itself be a very powerful *counting indistinguishability theorem*. The most general converse does not hold, but we prove the following, still highly general, version: if any two sets of real-valued signatures are Holant-indistinguishable, then they are equivalent up to an orthogonal transformation. This resolves a partially open conjecture of Xia (2010). Consequences of this theorem include the well-known result that homomorphism counts from all graphs determine a graph up to isomorphism, the classical sufficient condition for simultaneous orthogonal similarity of sets of real matrices, and a combinatorial characterization of simultaneously orthogonally decomposable (odeco) sets of symmetric tensors.

1 Introduction

Holant problems. Holant problems were introduced by Cai, Lu, and Xia [13] as a highly expressive framework for studying the computational complexity of counting problems. The problem Holant($\mathcal{F}$) is defined by a set $\mathcal{F}$ of *signatures*, where a signature F of *arity* n on *domain* $[q] := \{0, 1, \ldots, q-1\}$ is a tensor in $(\mathbb{C}^q)^{\otimes n}$, or equivalently a function $[q]^n \to \mathbb{C}$. Given a *signature grid* Ω – a

multigraph in which every degree-n vertex is assigned a n-ary signature from $\mathcal{F}$ – the problem is to compute the *Holant value* of Ω, which is the value of the contraction of Ω as a tensor network (see Section 2.1 for formal definitions). For various $\mathcal{F}$, Holant($\mathcal{F}$) captures a wide variety of natural counting problems on graphs, including counting partial or perfect matchings, graph homomorphisms, proper vertex or edge-colorings, or Eulerian orientations. Major complexity dichotomies classifying Holant($\mathcal{F}$) as either polynomial-time tractable or #P-hard, depending on $\mathcal{F}$, have been proved for various combinations of restrictions on $\mathcal{F}$ – for example, requiring that the signatures in $\mathcal{F}$ be real- or nonnegative-real-valued, symmetric (invariant under reordering of their inputs), or on the Boolean domain $q = 2$ [14, 16, 31, 32, 46].

Holant problems were motivated by Valiant's technique of *holographic transformations* [50]. In particular, Valiant's *Holant theorem* (Theorem 10 below) states roughly that two any signature sets $\mathcal{F}$ and $\mathcal{G}$ that are equivalent up to a certain linear transformation are *Holant-indistinguishable*, meaning that each signature grid Ω has the same Holant value whether its vertices are assigned signatures from $\mathcal{F}$ or from $\mathcal{G}$. Many problems which do not otherwise appear tractable are in fact tractable under a Holographic transformation to a known tractable problem [49]. Xia [53] conjectured the converse of the Holant theorem: if $\mathcal{F}$ and $\mathcal{G}$ are Holant-indistinguishable, then they are equivalent up to linear transformation. Xia's general conjecture is false [16], but one case highlighted by Xia was left open. This paper proves that case, which is as follows.

Theorem 1 (Theorem 12, informal)**.** *Let $\mathcal{F}$ and $\mathcal{G}$ be sets of real-valued signatures. Then $\mathcal{F}$ and $\mathcal{G}$ are equivalent under a real orthogonal transformation if and only if $\mathcal{F}$ and $\mathcal{G}$ are Holant-indistinguishable.*

Later, the techniques developed in a previous version of the present work [54] (including the direct sum and subdomain-restriction ideas in Lemmas 39 and 21 below) were used to show that Xia's general conjecture holds for any $\mathcal{F}$ and $\mathcal{G}$ which are *quantum-nonvanishing* [11]. The $\mathcal{F}$ and $\mathcal{G}$ in the context of Theorem 12 are quantum-nonvanishing, so a slightly weaker version of Theorem 12 in which the obtained orthogonal matrix is complex also follows from this more general converse.

Vertex and edge coloring models. This work uses and generalizes ideas from the theory of vertex coloring models and edge coloring models, two well-studied classes of Holant problems. De la Harpe and Jones [21] defined vertex and edge coloring models as extensions of statistical mechanics models (e.g. the Ising model), calling them "spin models" and "vertex models", respectively. A vertex coloring model (also called a spin system) is defined by a graph X with edge and possibly vertex weights. Given an input graph K, one aims to compute the *partition function*, the number of (weighted) graph homomorphisms from K to X. An edge coloring model is defined by a set $\mathcal{F}$ of symmetric signatures containing exactly one signature of each arity, and the problem of computing its partition function is equivalent to Holant($\mathcal{F}$) (this restriction on $\mathcal{F}$ ensures

that edge coloring models take ordinary graphs, rather than signature grids, as input).

One thread of prior work on vertex and edge coloring models characterizes which scalar-valued functions on graphs are expressible as vertex coloring models [28, 44] or as edge coloring models [25, 39, 42, 48]. Another, related, line of works compute the rank of *connection matrices* for vertex coloring models [35] and edge coloring models [24, 37]. See [38] for an overview of many of the above results. Following Freedman, Lovász, and Schrijver [28], these works use (labeled) *quantum graphs*, algebras of formal linear combinations of graphs equipped with labeled vertices or "half edges" incident to a single vertex, special cases of our *quantum gadgets* below. Many of these works also apply techniques from invariant theory, either of the symmetric group in the case of vertex coloring models [44], or, as in this work, of the orthogonal group $O(q)$ in the case of edge-coloring models [24, 25, 37, 39, 42, 48].

Counting Indistinguishability Theorems. Theorem 12 is a very general and powerful algebraic *counting indistinguishability theorem*. Such a theorem proves that two signatures, or sets of signatures, are indistinguishable as parameters for a counting problem if and only if they are equivalent under an algebraic transformation. These theorems exist for both vertex and edge coloring models, as well as other counting problems [1, 22, 26, 30, 40]. Since Holant captures a wide variety of counting problems, many such theorems are special cases of Theorem 12 (see section 5). If the counting problem in question is a generalized vertex coloring model (Holant($\mathcal{F} \cup \mathcal{EQ}$) for some $\mathcal{F}$), then the algebraic transformation is isomorphism, and if the counting problem is, as in this work, a generalized edge coloring model, then it is orthogonal. The first counting indistinguishability theorem, proved by Lovász [34], states that two graphs are isomorphic if and only if they admit the same number of homomorphisms from all graphs. Much later, Lovász [35] extended this theorem to vertex coloring models with nonnegative real weights, followed by extensions to complex edge weights by Schrijver [44], and to weights from any field of characteristic zero by Cai and Govorov [8]. Young [55] extended Cai and Govorov's proof to #CSP, or Holant($\mathcal{F} \cup \mathcal{EQ}$) for any $\mathcal{F}$.

For edge coloring models, Schrijver [42] showed that $\mathcal{F}$ and $\mathcal{G}$ define indistinguishable real edge coloring models if and only if $\mathcal{F}$ and $\mathcal{G}$ are equivalent under a real orthogonal transformation. This is a special case of our Theorem 12. Schrijver's proof exploits the specific nature of edge coloring models – that $\mathcal{F}$ and $\mathcal{G}$ consist of symmetric signatures and exactly one signature per arity – to transform input graphs into polynomials expressible in variables $y_1, \ldots, y_q$ (for $\mathcal{F}$ and $\mathcal{G}$ on domain $[q]$), where a monomial with variable multiset $\{y_{i_1}, \ldots, y_{i_n}\}$ corresponds to the $\{i_1, \ldots, i_n\}$-entry of the unique n-ary signature in $\mathcal{F}$. Another form of this result (in which the orthogonal matrix is allowed to be complex) follows from Regts' proof of [39, Lemma 5], which similarly encodes $\mathcal{F}$ and $\mathcal{G}$ as polynomials.

Mančinska and Roberson [36] showed that two graphs are *quantum iso-*

morphic – a relaxation of isomorphism originally defined in terms of quantum strategies for a nonlocal game [2] – if and only if they admit the same number of homomorphisms from all planar graphs. Cai and Young [10] extended this result to planar #CSP (or Pl-Holant($\mathcal{F} \cup \mathcal{EQ}$), where Pl-Holant restricts to planar signature grids).

The theory of counting indistinguishability is closely related to the computational complexity of counting problems, especially graph homomorphisms. The quantum graph algebras of Freedman, Lovász, and Schrijver [28] – which, along with their generalizations such as, in this work, quantum gadgets, have played central roles in many indistinguishability results [10, 30, 35, 36, 44] – have also been employed to study the parameterized complexity of subgraph counting [19, 20] and the complexity of counting homomorphisms from planar graphs [17]. The complexity of homomorphism indistinguishability itself – that is, of the problem of determining, for a fixed graph class $\mathfrak{G}$, whether input graphs X and Y admit the same number of homomorphisms from every graph in $\mathfrak{G}$, has also seen recent study [45]. Simultaneously, the complexity of the problem of determining the existence of invertible, including orthogonal, transformations between two tensors, called *tensor isomorphism* in this context, has also been investigated [18, 29]. The connection between indistinguishability and tensor isomorphism developed in this work is a first step towards developing connections between these previously disparate areas of complexity theory.

Odeco signature sets A real-valued symmetric signature (tensor) is *orthogonally decomposable*, or *odeco* [41], if it is orthogonally transformable to a signature in $\mathcal{GEQ}$, the set of *generalized equality* signatures, which take nonzero values only when all of their inputs are equal. Hence odeco tensors generalize diagonalizable matrices. Call a set $\mathcal{F}$ of signatures odeco if the signatures are simultaneously odeco (there is a single orthogonal transformation mapping $\mathcal{F}$ into $\mathcal{GEQ}$). In counting complexity, if $\mathcal{F}$ is odeco, then Holant($\mathcal{F}$) is polynomial-time tractable, as $\mathcal{F}$ maps into $\mathcal{GEQ}$, a trivially tractable set, under an orthogonal holographic transformation. Indeed, the tractability of *Fibonacci* signature sets [12] can, with one exception, be explained by such sets being simultaneously odeco (see e.g. [7, Section 2.2]). Fibonacci sets constitute almost all nontrivial tractable cases of Holant* problems (an important variant of Holant in which all unary signatures are assumed present) for symmetric signatures on the Boolean domain [13]. Simultaneously odeco sets provide a natural starting point for extending Fibonacci signatures to higher domains [9, 33], where no full complexity dichotomy for Holant* is known.

Boralevi, Draisma, Horobeţ, and Robeva [4], resolving a conjecture of Robeva [41], showed using techniques from algebraic geometry that a single tensor F is odeco if and only if the signature of a certain F-gadget is symmetric. Using Theorem 12, we in Theorem 49 extend this characterization to sets of simultaneously odeco signatures: $\mathcal{F}$ is odeco if and only if every connected $\mathcal{F}$-gadget has a symmetric signature. The latter condition is equivalent to the symmetry of the signatures of a set of small gadgets constructed from every pair of

signatures in $\mathcal{F}$. Therefore, if $\mathcal{F}$ is finite, our characterization yields a simple $O(q^{2n-2}|\mathcal{F}|^2)$-time algorithm (for $\mathcal{F}$ on domain $[q]$ with maximum arity n) for deciding whether $\mathcal{F}$ is odeco. Our characterization also deepens the connection between Fibonacci and odeco signatures, as the original proof of tractability of any Fibonacci signature set $\mathcal{F}$ [12] relied on the fact that every connected $\mathcal{F}$-gadget has a signature which is itself Fibonacci (in particular, is symmetric). One can view the (iii) $\implies$ (ii) result in Theorem 49 as a general-domain version of this proof.

Overview. We introduce the necessary preliminaries and formally state the main theorem, Theorem 12, in section 2. We prove Theorem 12 in Sections 3 and 4. The latter section 4 is mostly algebraic in nature. It applies an invariant-theoretic result of Schrijver [43] to prove an algebraic characterization (Theorem 37) of quantum gadgets, which we call *duality*. The proof of this characterization follows and generalizes a proof of a similar result of Regts [37] for edge coloring models. By unifying the perspective of Regts with that of Mančinska and Roberson and Cai and Young [10, 36, 55] we find our proof analogous to proofs of similar results in the latter line of work (see Remark 38). The former section 3 is mostly combinatorial in nature, and uses the duality theorem of the section 4 as a black box. While this duality is analogous to those used by Mančinska and Roberson and Cai and Young, the proofs of their ensuing counting indistinguishability theorems use the discrete *orbits* of the action of the symmetric or quantum symmetric group on the domain set $[q]$, which do not exist for the continuous group $O(q)$ of orthogonal transformations in our setting. Instead, we apply a novel method: induction on the domain size q. We show that, if $\mathcal{F}$ and $\mathcal{G}$ contain a binary signature represented by a nontrivial (i.e., not a multiple of I) diagonal matrix D, then we can separate $[q]$ into smaller subdomains and, for each such subdomain $X \subset [q]$, inductively obtain an orthogonal transformation between the restrictions $\mathcal{F}|_X$ and $\mathcal{G}|_X$ of $\mathcal{F}$ and $\mathcal{G}$ to X, then combine these into a full orthogonal transformation between $\mathcal{F}$ and $\mathcal{G}$. We then use duality to nonconstructively obtain this matrix D.

In section 5, we show that Theorem 12 encompasses a wide range of existing counting indistinguishability theorems, and yields some novel variations of these theorems. We also prove our combinatorial characterization of odeco signature sets. Finally, in section 6, we conjecture a variation of Theorem 12 extending the results of Mančinska and Roberson [36] and Cai and Young [10] to planar-Holant-indistinguishability and quantum orthogonal transformations.

2 Preliminaries, Background, and the Main Theorem

2.1 Holant Problems

A *multigraph* $K = (V(K), E(K))$ is defined by a finite set $V(K)$ of vertices and a multiset $E(K)$ of edges, which are one- or two-element subsets of $V(K)$. That

is, edges are undirected and loops are allowed. A *graph* K has no loops and at most one edge between every pair of vertices.

A *signature* F of finite *arity* $n \geq 0$ on finite *domain* $V(F)$ is a function $V(F)^n \to \mathbb{R}$. We will often take $V(F) = [q] := \{0, 1, \ldots, q-1\}$, in which case we also view F as a tensor in $(\mathbb{R}^q)^{\otimes n}$. For $\mathbf{x} = (x_1, \ldots, x_n) \in V(F)^n$, abbreviate $F_{\mathbf{x}} := F(x_1, \ldots, x_n) \in \mathbb{R}$. Signature F is *symmetric* if its value depends only on the multiset of inputs, not on their order. Any time we consider a set $\mathcal{F}$ of signatures, we assume that all signatures in $\mathcal{F}$ have the same domain, denoted $V(\mathcal{F})$.

Definition 2 ($T(\mathbb{R}^q)$)**.** *Following Schrijver [43], define*

$$T(\mathbb{R}^q) := \bigcup_{n \geq 0} (\mathbb{R}^q)^{\otimes n}$$

to be the set of all real-valued signatures on domain $[q]$*.*

In the context of a signature set $\mathcal{F}$, a *signature grid* (or $\mathcal{F}$*-grid*) Ω consists of an underlying multigraph with vertex set V and edge set E, and an assignment of an $\deg(v)$-ary signature $F_v \in \mathcal{F}$ to each $v \in V$, along with an ordering of the edges $\delta(v) \subset E$ incident to v to serve as the $\deg(v)$ input variables to F. That is, there is an ordering $e_1, \ldots, e_{\deg(v)}$ of v's incident edges such that, if $\sigma : E \to V(\mathcal{F})$ is an assignment of a value in $V(\mathcal{F})$ to each edge of Ω, then F_v evaluates to $F_v(\sigma|_{\delta(v)}) := F_v(\sigma(e_1), \ldots, \sigma(e_{\deg(v)}))$. Somewhat unusually, we also allow E to contain *vertexless loops*, edges whose two ends are connected to each other, with no incident vertices. The problem Holant($\mathcal{F}$) is defined as follows: given an $\mathcal{F}$-grid Ω with vertex set V and edge set E, compute the *Holant value*

$$\text{Holant}_\Omega(\mathcal{F}) := \sum_{\sigma : E \to V(\mathcal{F})} \prod_{v \in V} F_v(\sigma|_{\delta(v)}). \tag{1}$$

When $\mathcal{F}$ is clear from context, we abbreviate $\text{Holant}_\Omega(\mathcal{F})$ as Holant_Ω. Each vertexless loop of Ω contributes a global factor $|V(\mathcal{F})|$ to Holant_Ω. More generally, the Holant value of a disconnected signature grid is the product of the Holant values of its connected components. For signature sets $\mathcal{F}$ and $\mathcal{F}'$, define Holant($\mathcal{F} \mid \mathcal{F}'$) as the Holant problem whose input is a bipartite $(\mathcal{F} \sqcup \mathcal{F}')$-grid Ω ($\sqcup$ denotes disjoint union), with the vertices in the two bipartitions assigned signatures in $\mathcal{F}$ and $\mathcal{F}'$, respectively.

For example, if $\mathcal{F}$ is on the Boolean domain $\{0, 1\}$ and consists of, for each arity n, a symmetric signature which evaluates to 1 on input strings of Hamming weight 1 and 0 on all other input strings, then $\text{Holant}_\Omega(\mathcal{F})$ equals the number of perfect matchings in the multigraph underlying Ω. For another example, consider counting graph homomorphisms from a multigraph K to a graph X, where a graph homomorphism from K to X is a map $\varphi : V(K) \to V(X)$ such that $uv \in E(K) \implies \varphi(u)\varphi(v) \in V(X)$. Let $A_X \in \{0, 1\}^{q \times q}$ be the adjacency matrix of q-vertex graph X, and define the set $\mathcal{EQ} = \{=_n \mid n \geq 1\}$ of *equality* signatures on domain $[q]$, where $=_n (x_1, \ldots, x_n)$ is 1 if $x_1 = \ldots = x_n$, and is 0 otherwise. Consider a Holant($A_X \mid \mathcal{EQ}$) grid Ω. We can think of any edge

assignment σ with a nonzero contribution to the Holant value as a map from vertices in Ω assigned $\mathcal{EQ}$ signatures to values in $[q]$, or equivalently vertices of X. This map must send the two $\mathcal{EQ}$ vertices adjacent to every A_X vertex to an edge in A_X, so, if K_Ω is the multigraph resulting from ignoring (i.e. treating as edges) the degree-two vertices assigned A_X in the underlying multigraph of Ω, then $\text{Holant}_\Omega(A_X \mid \mathcal{EQ})$ equals the number of homomorphisms from K_Ω to X. See Figure 1.

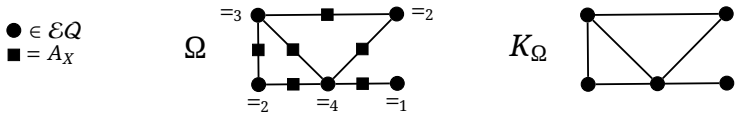


Figure 1: An $(A_X \mid \mathcal{EQ})$-grid Ω and the multigraph K_Ω such that $\text{Holant}_\Omega(A_X \mid \mathcal{EQ})$ counts the number of homomorphisms from K_Ω to X.

2.2 Gadgets and Signature Matrices

Instead of viewing a signature F as a tensor in $(\mathbb{R}^q)^{\otimes n}$ or function in $\mathbb{R}^{[q]^n}$, we can partition its inputs in two to view it naturally as a matrix.

Definition 3 $(F^{m,d}, f)$**.** *For $F \in (\mathbb{R}^q)^{\otimes n}$ and $m, d \geq 0$ with $m + d = n$, define the* (m, d)-signature matrix*, or* flattening*, $F^{m,d} \in \mathbb{R}^{q^m \times q^d}$ of F by, for $\mathbf{x} \in [q]^m$ and $\mathbf{y} \in [q]^d$,*

$$(F^{m,d})_{\mathbf{x},\mathbf{y}} = F(x_0, \ldots, x_{m-1}, y_{d-1}, \ldots, y_0),$$

where we use the standard isomorphism $[q^n] \cong [q]^n$ to index $F^{m,d}$.

Abbreviate $f = F^{n,0} \in \mathbb{R}^{q^n}$ – the (column) signature vector *of F.*

We will often identify binary signatures in $(\mathbb{R}^q)^{\otimes 2}$ with their $1, 1$ signature matrices in $\mathbb{R}^{q \times q}$.

Definition 4 (Gadget, $\mathfrak{G}_\mathcal{F}, \mathfrak{G}_\mathcal{F}(m, d)$)**.** *For signature set $\mathcal{F}$, an $\mathcal{F}$-*gadget *$\mathbf{K}$ is identical to an $\mathcal{F}$-grid, but with an enhanced edge set $E(\mathbf{K}) = E'(\mathbf{K}) \cup L(\mathbf{K}) \cup R(\mathbf{K})$, where $E'(\mathbf{K})$ is the usual set of $\mathcal{F}$-grid* internal *edges and $L(\mathbf{K})$ and $R(\mathbf{K})$ are ordered lists of left and right* dangling edges*, each specified by either a single endpoint in $V(\mathbf{K})$, or paired with another element of $L(\mathbf{K})$ or $R(\mathbf{K})$ as a two-sided dangling edge – a* wire *– with no incident vertices.*

Define $\mathfrak{G}_\mathcal{F}$ to be the set of all $\mathcal{F}$-gadgets, and $\mathfrak{G}_\mathcal{F}(m, d) \subset \mathfrak{G}_\mathcal{F}$ to be the set of gadgets with m left and d right dangling edges. The m left dangling edges $\ell_0, \ldots, \ell_{m-1}$ and d right dangling edges $r_0, \ldots, r_{d-1}$ are drawn on the left and right of $\mathbf{K}$, respectively, with dangling ends (the ends with no endpoint) from top to bottom, so in overall counterclockwise cyclic order $\ell_0, \ldots, \ell_{m-1}, r_{d-1}, \ldots, r_0$ around $\mathbf{K}$. The two dangling ends of a wire are part of the same dangling edge, so $\ell_0, \ldots, \ell_{m-1}, r_{d-1}, \ldots, r_0$ may not be distinct.

See Figure 2 for examples of gadgets. Connecting the two ends of a wire, we obtain a vertexless loop. A gadget in $\mathfrak{G}_{\mathcal{F}}(m,d)$ defines a $(m+d)$-ary signature in flattened form, with dangling edges representing inputs, as follows (cf. (1)):

Definition 5 ($M(\mathbf{K})$)**.** *Define the* signature matrix $M(\mathbf{K}) \in \mathbb{R}^{q^m \times q^d}$ *of* $\mathbf{K} \in \mathfrak{G}_{\mathcal{F}}(m,d)$ *by*

$$M(\mathbf{K})_{\mathbf{x},\mathbf{y}} = \sum_{\substack{\sigma: E(\mathbf{K}) \to [q] \\ \forall i: \sigma(\ell_i) = x_i \\ \forall j: \sigma(r_j) = y_j}} \prod_{v \in V} F_v(\sigma|_{\delta(v)}) \quad \text{for } \mathbf{x} \in [q]^m \text{ and } \mathbf{y} \in [q]^d.$$

In other words, $M(\mathbf{K})_{\mathbf{x},\mathbf{y}}$ equals the Holant value of $\mathbf{K}$ when the left and right dangling edges are fixed to the domain elements in $\mathbf{x}$ and $\mathbf{y}$, respectively. As a gadget with one left and one right dangling end, a wire has siganture matrix I (the identity), as its left and right inputs must agree. If $\mathbf{K} \in \mathfrak{G}_{\mathcal{F}}(m,d)$ is a gadget consisting of a single vertex assigned F with m and d incident left and right dangling edges, respectively, ordered to match the input order of F, then $M(\mathbf{K}) = F^{m,d}$. In general, for $\mathbf{K} \in \mathfrak{G}_{\mathcal{F}}(m,d)$, there is a unique $F \in (\mathbb{R}^q)^{\otimes m+d}$, called the *signature of* $\mathbf{K}$, such that $M(\mathbf{K}) = F^{m,d}$. Note that F does not depend on the particular left/right partition (i.e. choice of m and d) of a fixed cyclic order of $\mathbf{K}$'s dangling edges.

Definition 6 ($\circ, \otimes, \top$)**.** *Define the following three operations on gadgets:*

- *For* $\mathbf{K} \in \mathfrak{G}_{\mathcal{F}}(m,d)$ *and* $\mathbf{L} \in \mathfrak{G}_{\mathcal{F}}(d,r)$*, construct* $\mathbf{K} \circ \mathbf{L} \in \mathfrak{G}_{\mathcal{F}}(m,r)$ *by connecting the dangling ends of* $r_i \in R(\mathbf{K})$ *and* $\ell_i \in L(\mathbf{L})$ *for* $i \in [d]$*.*
- *For* $\mathbf{K} \in \mathfrak{G}_{\mathcal{F}}(m_1,d_1)$ *and* $\mathbf{L} \in \mathfrak{G}_{\mathcal{F}}(m_2,d_2)$*, construct* $\mathbf{K} \otimes \mathbf{L} \in \mathfrak{G}_{\mathcal{F}}(m_1+m_2, d_1+d_2)$ *as the disjoint union of* $\mathbf{K}$ *and* $\mathbf{L}$*, placing* $\mathbf{K}$ *above* $\mathbf{L}$*. Interleave* $\mathbf{K}$ *and* $\mathbf{L}$*'s dangling edges into an overall cyclic order:* $\mathbf{K}$ *left, then* $\mathbf{L}$ *left, then* $\mathbf{L}$ *right, then* $\mathbf{K}$ *right.*
- *For* $\mathbf{K} \in \mathfrak{G}_{\mathcal{F}}(m,d)$*, construct* $\mathbf{K}^\top \in \mathfrak{G}_{\mathcal{F}}(d,m)$ *by exchanging the roles of left and right dangling edges and reversing the overall cyclic dangling edge order (visually, horizontally reflect* $\mathbf{K}$*).*

See Figure 2. As one might expect, the gadget operations $\circ, \otimes, \top$ induce the respective operations – composition, Kronecker product, transpose – on their signature matrices. See e.g. [7, Section 1.3].

Definition 7 ($\langle \cdot, \cdot \rangle, \| \cdot \|$)**.** *For real-valued* $\mathcal{F}$ *and* n*-ary gadgets* $\mathbf{K}$ *and* $\mathbf{L}$*, construct the signature grid* $\langle \mathbf{K}, \mathbf{L} \rangle$ *by connecting the* i*th dangling edges of* $\mathbf{K}$ *and* $\mathbf{L}$*, for* $i \in [n]$*. If* $\mathbf{K}$ *and* $\mathbf{L}$ *have signatures* K *and* L*, then define* $\langle K, L \rangle := \text{Holant}_{\langle \mathbf{K}, \mathbf{L} \rangle} = \langle K^{n,0}, L^{n,0} \rangle$ *(the standard inner product on* $\mathbb{R}^{q^n}$*). Define* $\|F\| := \sqrt{\langle F, F \rangle} = \sqrt{\sum_{\mathbf{x}} F_{\mathbf{x}}^2}$*.*

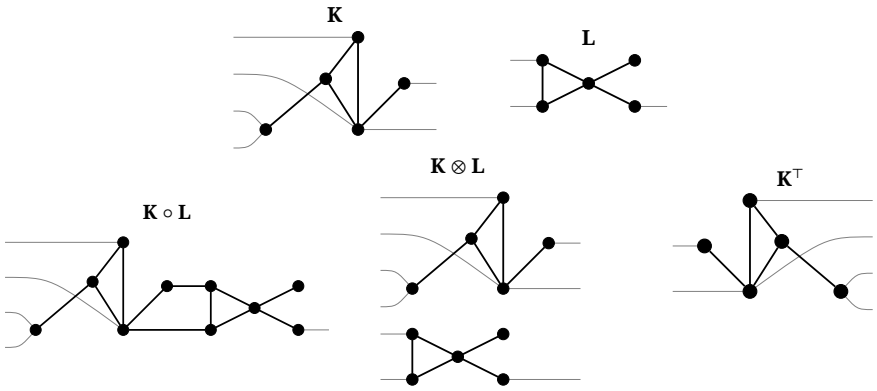


Figure 2: Operations on gadgets $\mathbf{K} \in \mathfrak{G}(4,2)$ and $\mathbf{L} \in \mathfrak{G}(2,1)$. Dangling edges are drawn thinner than internal edges.

2.3 The Holant Theorem

Throughout this work, we will be considering pairs of multisets of signatures, called simply *signature sets*, usually denoted $\mathcal{F}$ and $\mathcal{G}$. Such pairs are assumed to be *similar*, meaning they have the same domain size q and they come equipped with a multiset bijection between $\mathcal{F}$ and $\mathcal{G}$ such that, for n-ary $F \in \mathcal{F}$, the image $G \in \mathcal{G}$ of F, called the signature *corresponding* to F, has the same arity n. This correspondence bijection, usually given implicitly along with a similar pair $\mathcal{F}, \mathcal{G}$, is denoted by $\leftrightsquigarrow$. That is, $\mathcal{F} \ni F \leftrightsquigarrow G \in \mathcal{G}$ indicates that G corresponds to F. We consider multisets, rather than sets, of signatures because we will wish to preserve the bijective correspondence between $\mathcal{F}$ and $\mathcal{G}$ when adding signatures to $\mathcal{F}$ and $\mathcal{G}$ that may already be present.

Definition 8 $(\mathbf{K}_{\mathcal{F}\to\mathcal{G}}, \Omega_{\mathcal{F}\to\mathcal{G}})$**.** *For similar signature sets $\mathcal{F}$ and $\mathcal{G}$ with correspondence bijection $\leftrightsquigarrow$ and gadget $\mathbf{K} \in \mathfrak{G}_{\mathcal{F}}$, define the gadget $\mathbf{K}_{\mathcal{F}\to\mathcal{G}} \in \mathfrak{G}_{\mathcal{G}}$ by replacing every $F \in \mathcal{F}$ assigned to a vertex in $\mathbf{K}$ with the $G \in \mathcal{G}$ such that $F \leftrightsquigarrow G$. If $\mathbf{K}$ has zero dangling edges then it is an $\mathcal{F}$-grid Ω, and is transformed to a $\mathcal{G}$-grid $\Omega_{\mathcal{F}\to\mathcal{G}}$.*

We next introduce the concept of signature transformation.

Definition 9 $(HF, H\,\mathcal{F})$**.** *Define an action of the group $GL_q(\mathbb{R})$ of invertible $q \times q$ matrices on the set $T(\mathbb{R}^q)$ of domain-$[q]$ signatures as follows: for $H \in GL_q(\mathbb{R})$ and $F \in (\mathbb{R}^q)^{\otimes n}$ with signature vector f, let $HF \in (\mathbb{R}^q)^{\otimes n}$ be the signature whose signature vector is $H^{\otimes n} f$ – that is, $(HF)^{n,0} = H^{\otimes n} f$ (see Figure 3).*

For a signature set $\mathcal{F} \subset T(\mathbb{R}^q)$, define $H\,\mathcal{F} := \{HF \mid F \in \mathcal{F}\}$.

Holant problems were originally motivated by the following theorem, a powerful reduction tool proved by Valiant [50]. For $F \in (\mathbb{R}^q)^{\otimes n}$ and $A \in \mathrm{GL}_q$, define FA similarly to AF, by $(FA)^{0,n} = F^{0,n} A^{\otimes n}$.

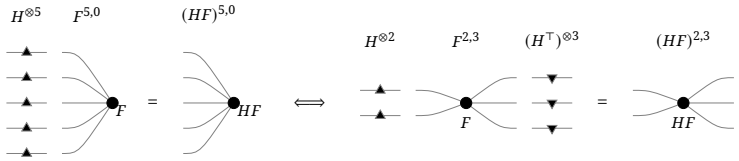


Figure 3: The gadget perspective on $H^{\otimes n} f = (HF)^{n,0}$, or equivalently $H^{\otimes m} F^{m,d} (H^{\top})^{\otimes d} = (HF)^{m,d}$. Each triangle is a vertex assigned H as a binary signature.

Theorem 10 (The Holant Theorem)**.** *For any $(\mathcal{F} \mid \mathcal{F}')$-grid Ω and matrix $A \in GL_q$,*

$$\text{Holant}_{\Omega}(\mathcal{F} \mid \mathcal{F}') = \text{Holant}_{\Omega'}(A\,\mathcal{F} \mid \mathcal{F}'\,A^{-1}),$$

where $\Omega' = \Omega_{(\mathcal{F} \mid \mathcal{F}') \to (A\,\mathcal{F} \mid \mathcal{F}'\,A^{-1})}$.

Xia [53] conjectured that the converse of Theorem 10 holds as long as $\mathcal{F}$ or $\mathcal{F}'$ contains a signature with arity greater than one – that is, if $\text{Holant}_{\Omega}(\mathcal{F} \mid \mathcal{F}') = \text{Holant}_{\Omega_{(\mathcal{F} \mid \mathcal{F}') \to (\mathcal{G} \mid \mathcal{G}')}}(\mathcal{G} \mid \mathcal{G}')$ for every $(\mathcal{F} \mid \mathcal{F}')$-grid Ω, then there is an $A \in \text{GL}_q$ such that $\mathcal{G} = A\,\mathcal{F}$ and $\mathcal{G}' = \mathcal{F}'\,A^{-1}$. However, Cai, Guo, and Williams [16, Section 4.3] observe that this conjecture is false. They consider the problems

$$\text{Holant}([0,1,0] \mid [a,b,1,0,0]) \quad \text{and} \quad \text{Holant}([0,1,0] \mid [0,0,1,0,0])$$

for any a, b not both 0, where $[0,1,0]$ and $[a,b,1,0,0]$ are symmetric signatures on the Boolean domain $\{0,1\}$ of arity $n = 2, 4$, respectively, specified by their values on input strings of Hamming weight 0 through n. For both problems, $[0,1,0]$ is only nonzero on edge assignment σ if σ assigns its two incident edges opposite values. Thus, unless its contribution to the total Holant value is 0, σ must assign 0 to exactly half the edges in Ω, and 1 to the other half. Then, since every $[a,b,1,0,0]$ evaluates to zero whenever it receives more 1 inputs than 0 inputs, no nonzero assignment σ assigns any $[a,b,1,0,0]$ more 0 inputs than 1 inputs. Therefore, in this context, $[a,b,1,0,0]$ is indistinguishable from $[0,0,1,0,0]$, so the hypothesis of Xia's conjecture is satisfied. However, there is no $A \in \text{GL}_2$ satisfying $A[0,1,0] = [0,1,0]$ and $[a,b,1,0,0]A^{-1} = [0,0,1,0,0]$. To see this, if $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, then, with the signature vector of $[0,1,0]$ being $(0,1,1,0)^{\top}$,

$$A^{\otimes 2} \begin{bmatrix} 0 \\ 1 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} a^2 & ab & ba & b^2 \\ ac & ad & bc & bd \\ ca & cb & da & db \\ c^2 & cd & dc & d^2 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 1 \\ 0 \end{bmatrix} \iff ab = cd = 0 \text{ and } ad{+}bc = 1 \implies \begin{cases} a = d = 0 \\ b = c = 0 \end{cases} \text{or}.$$

If $b = c = 0$ then A simply rescales a signature's entries, and if $a = d = 0$ then A rescales a signature's entries and exchanges the roles of 0 and 1, which has the effect of reversing the entries in the $[\cdot, \ldots, \cdot]$ notation. Thus $[0,0,1,0,0]A = [0,0,*,0,0] \neq [a,b,1,0,0]$.

This counterexample exists due to the bipartiteness enforced by Holant($\mathcal{F} \mid \mathcal{G}$), because

$$\langle [0,0,1,0,0], [0,0,1,0,0] \rangle = \|[0,0,1,0,0]\|^2 \neq \|[a,b,1,0,0]\|^2 = \langle [a,b,1,0,0], [a,b,1,0,0] \rangle ,$$

so $[a,b,1,0,0]$ and $[0,0,1,0,0]$ are not indistinguishable on general (non-bipartite) signature grids. To avoid such counterexamples, we consider the following well-known form of Theorem 10 that applies to non-bipartite grids. An edge in an $\mathcal{F}$-grid Ω, viewed on its own, is a wire, with signature I. Therefore we can, without changing the Holant value, replace each edge in Ω by a binary gadget with a single vertex assigned I, splitting the edge into a path of length two. Then Ω is a signature grid in the context of Holant($\mathcal{F} \mid I$), so, by Theorem 10, for any invertible H, $\text{Holant}_{\Omega}(\mathcal{F} \mid I) = \text{Holant}_{\Omega'}(H\mathcal{F} \mid \{I\}H^{-1})$. Let $O(q)$ be the group of $q \times q$ orthogonal matrices. If $H \in O(q)$ then, as in Figure 3, $(I^{2,0}(H^{-1})^{\otimes 2})^{1,1} = (H^{-1})^{\top} I H^{-1} = H H^{-1} = I$. Thus $\{I\}H^{-1} = \{I\}$ (see also Theorem 28 below), so $\text{Holant}_{\Omega}(\mathcal{F} \mid I) = \text{Holant}_{\Omega'}(H\mathcal{F} \mid I)$. Then Holant($H\mathcal{F} \mid I$) is again equivalent to Holant($H\mathcal{F}$), so we have proved the following.

Corollary 11 (The Orthogonal Holant Theorem)**.** *If $\mathcal{G} = H\mathcal{F}$ for orthogonal H, then $\text{Holant}_{\Omega}(\mathcal{F}) = \text{Holant}_{\Omega_{\mathcal{F}\to\mathcal{G}}}(\mathcal{G})$ for every $\mathcal{F}$-grid Ω.*

Xia [53] also considers the converse of Corollary 11, and proves that it holds for specific $\mathcal{F}$ and $\mathcal{G}$ consisting of real-valued symmetric signatures with small domain and/or arity. The main result of this work is the converse of Corollary 11 for any sets $\mathcal{F}$ and $\mathcal{G}$ of real-valued signatures, with no restrictions.

Theorem 12 (Main Result)**.** *The following are equivalent for similar sets $\mathcal{F}$, $\mathcal{G}$ of real-valued signatures.*

(i) $\text{Holant}_{\Omega}(\mathcal{F}) = \text{Holant}_{\Omega_{\mathcal{F}\to\mathcal{G}}}(\mathcal{G})$ *for every $\mathcal{F}$-grid Ω.*

(ii) There is a real orthogonal matrix H such that $H\mathcal{F} = \mathcal{G}$.

Call $\mathcal{F}$ and $\mathcal{G}$ satisfying (i) *Holant-indistinguishable*, and call $\mathcal{F}$ and $\mathcal{G}$ satisfying (ii) *ortho-equivalent*. The following two properties follow directly from the definitions and Corollary 11, but, as they will prove useful throughout the proof of Theorem 12, we state them explicitly.

Proposition 13. *For any orthogonal H_1 and H_2, $\mathcal{F}$ and $\mathcal{G}$ are ortho-equivalent/Holant-indistinguishable if and only if $H_1\mathcal{F}$ and $H_2\mathcal{G}$ are ortho-equivalent/Holant-indistinguishable, respectively.*

Proposition 14. *For any similar pair $\mathcal{F}$,$\mathcal{G}$ and additional similar pair $\mathcal{F}'$,$\mathcal{G}'$ of signature sets, the similarity bijections between $\mathcal{F}, \mathcal{G}$ and between $\mathcal{F}', \mathcal{G}'$ induce a similarity bijection between the disjoint unions $\mathcal{F} \sqcup \mathcal{F}'$ and $\mathcal{G} \sqcup \mathcal{G}'$, and if $\mathcal{F} \sqcup \mathcal{F}'$ and $\mathcal{G} \sqcup \mathcal{G}'$ are ortho-equivalent, then $\mathcal{F}$ and $\mathcal{G}$ are ortho-equivalent.*

Remark 15. *Although we will focus on real-valued signatures, the general and orthogonal Holant Theorems hold for complex-valued signatures. However, Theorem 12 does not hold for general sets $\mathcal{F}$ and $\mathcal{G}$ of complex-valued signatures, even when we allow H to be complex. For example, Draisma and Regts [24] consider the* vanishing *unary signature $F \in \mathbb{C}^{[2]^1}$ defined by $F_0 = 1$ and $F_1 = i$. Any F-grid Ω with at least one vertex satisfies $\text{Holant}_\Omega(F) = 0$, as Ω is a disjoint union of K_2 complete graphs, with each component having value $f^\top f = [1, i]^\top [1, i] = 0$. Thus F is Holant-indistinguishable from 0, but there is no orthogonal matrix H, real or complex, satisfying $Hf = H[1, i]^\top = [0, 0]^\top$. However, as we will discuss briefly in Section 6.1, Theorem 12 holds for complex $\mathcal{F}$ and $\mathcal{G}$ under certain additional assumptions.*

2.4 Block Matrices and Signatures

Definition 16. *Let $\mathcal{I}$ be an index/domain set, and $X \sqcup Y = \mathcal{I}$ be a nontrivial partition of $\mathcal{I}$.*

1. *For a matrix $H \in \mathbb{R}^{\mathcal{I} \times \mathcal{I}}$ and $R, C \in \{X, Y\}$, let $H|_{R,C} \in \mathbb{R}^{R \times C}$ be the submatrix of H with rows indexed by R and columns indexed by C. Up to row and column reordering, H is the block matrix*

$$H = \begin{bmatrix} H|_{X,X} & H|_{X,Y} \\ H|_{Y,X} & H|_{Y,Y} \end{bmatrix}.$$

2. *More generally, for a signature/tensor $F \in \mathbb{R}^{\mathcal{I}^n}$ and $\mathbf{R} \in \{X, Y\}^n$, define $F|_{\mathbf{R}} \in \mathbb{R}^{\mathbf{R}}$ (where we identify $\mathbf{R}$ with the set $\prod_{i=1}^n R_i$) to be the subtensor of F with ith input restricted to R_i. For signature set $\mathcal{F}$, let $\mathcal{F}\,|_{\mathbf{R}} := \{F|_{\mathbf{R}} : F \in \mathcal{F}\}$.*

   Abbreviate $F|_X := F|_{X^n}$ and $\mathcal{F}\,|_X := \mathcal{F}\,|_{X^n}$.

3. *Let $F^{m,d} \in \mathbb{R}^{\mathcal{I}^m \times \mathcal{I}^d}$ be a signature matrix, and let $\mathbf{R} \in \{X, Y\}^m$ and $\mathbf{C} \in \{X, Y\}^d$. Define $F^{m,d}|_{\mathbf{R},\mathbf{C}} \in \mathbb{R}^{\mathbf{R} \times \mathbf{C}}$ as the submatrix of $F^{m,d}$ with rows indexed by $\mathbf{R}$ and columns indexed by $\mathbf{C}$ (in other words, $F^{m,d}|_{\mathbf{R},\mathbf{C}} = (F|_{\mathbf{R}\,\mathbf{C}})^{m,d}$, where $\mathbf{R}\,\mathbf{C}$ is the concatenation of $\mathbf{R}$ and $\mathbf{C}$).*

Definition 17 ($\oplus$)**.** *Let F, G be n-ary signatures on domains $V(F)$, $V(G)$, both of size q. The* direct sum *$F \oplus G$ of F and G is an n-ary signature on domain $V(F) \sqcup V(G)$ defined by*

$$(F \oplus G)_{\mathbf{x}} = \begin{cases} F_{\mathbf{x}} & \mathbf{x} \in V(F)^n \\ G_{\mathbf{x}} & \mathbf{x} \in V(G)^n \\ 0 & \text{otherwise} \end{cases} \qquad \text{for } \mathbf{x} \in (V(F) \sqcup V(G))^n.$$

For signature sets $\mathcal{F}$ and $\mathcal{G}$, define $\mathcal{F} \oplus \mathcal{G} = \{F \oplus G \mid \mathcal{F} \ni F \rightsquigarrow G \in \mathcal{G}\}$.

We will frequently apply Definition 16 to the partition $V(\mathcal{F}) \sqcup V(\mathcal{G})$ of the domain of $\mathcal{F} \oplus \mathcal{G}$. It follows directly from the definitions that, for $\mathcal{F} \ni F \leftrightsquigarrow G \in \mathcal{G}$ and any $m + d = \text{arity}(F)$,

$$(F \oplus G)^{m,d}|_{\mathbf{R},\mathbf{C}} = \begin{cases} F^{m,d} & \mathbf{R} = V(\mathcal{F})^m \wedge \mathbf{C} = V(\mathcal{F})^d \\ G^{m,d} & \mathbf{R} = V(\mathcal{G})^m \wedge \mathbf{C} = V(\mathcal{G})^d \\ 0 & \text{otherwise.} \end{cases} \tag{2}$$

Appendix A states and proves some additional necessary results involving actions of block matrices on block signatures. These additional results confirm that these block structures interact as one would expect, analogously to ordinary block matrix algebra.

3 Domain Induction: The Proof of Theorem 12

Throughout the proof of Theorem 12, we would like to assume that $\mathcal{F}$ and $\mathcal{G}$ contain not only the signatures of all $\mathcal{F}$ and $\mathcal{G}$-gadgets, but all linear combinations of such signatures. There are two principal reasons for this. First, the invariant-theoretic machinery in section 4 demands that our signature sets be vector spaces closed under $\circ, \otimes, \top$. Second, for the domain induction argument we will need $\mathcal{F}$ and $\mathcal{G}$ to contain the matrices (binary signatures) projecting onto certain subdomains. We obtain these projectors via polynomial interpolation, which requires both matrix composition (gadget composition) and linear combinations. These considerations motivate the following definition.

Definition 18 $(\mathfrak{Q}_{\mathcal{F}}(m,d), [\mathcal{F}])$**.** *A* (m,d)-quantum $\mathcal{F}$-gadget *is a formal (finite)* $\mathbb{R}$*-linear combination of gadgets in* $\mathfrak{G}_{\mathcal{F}}(m,d)$*. Let* $\mathfrak{Q}_{\mathcal{F}}(m,d)$ *be the set of all* (m,d)*-quantum* $\mathcal{F}$*-gadgets, and* $\mathfrak{Q}_{\mathcal{F}} = \bigcup_{m,d} \mathfrak{Q}_{\mathcal{F}}(m,d)$*.*

Extend the signature matrix function M *linearly to* $\mathfrak{Q}_{\mathcal{F}}$*. Then define the* quantum gadget closure $[\mathcal{F}]$ *of* $\mathcal{F}$ *as the multiset of signatures obtainable from* $\mathcal{F}$ *by quantum gadget construction:*

$$[\mathcal{F}] = \bigsqcup_{\mathbf{K} \in \mathfrak{Q}_{\mathcal{F}}} (\textit{signature of } \mathbf{K}).$$

In particular, $[\mathcal{F}]$ is in bijection with $\mathfrak{Q}_{\mathcal{F}}$, so, if $\mathcal{F}$ and $\mathcal{G}$ are similar, then $[\mathcal{F}]$ and $[\mathcal{G}]$ are similar, with the signature of $\mathbf{K} \in \mathfrak{Q}_{\mathcal{F}}$ corresponding to the signature of $\mathbf{K}_{\mathcal{F}\to\mathcal{G}} \in \mathfrak{Q}_{\mathcal{G}}$. A signature set $\mathcal{F} \subset T(\mathbb{R}^q)$ is *quantum-gadget-closed* if $\mathcal{F} = [\mathcal{F}]$ as sets.

The proof of Theorem 12 now proceeds roughly as follows. First, Lemma 19 lets us replace $\mathcal{F}$ and $\mathcal{G}$ by their quantum gadget closures. Then we use the invariant-theoretic duality theory from section 4, in the form of Lemma 25, to obtain a nonzero diagonal matrix D *intertwining* (roughly, non-invertibly transforming between) $\mathcal{F}$ and $\mathcal{G}$. If D is a scalar multiple of the identity matrix, then this multiple must be ± 1 and D is the desired orthogonal transformation. Otherwise, from D we interpolate the projectors onto disjoint eigenspaces of D

– two subdomains forming a nontrivial partition of the full domain $[q]$. These projectors pass indistinguishability to the two subdomains (Lemma 21). Then, by induction on the domain size, we obtain orthogonal transformations between $\mathcal{F}$ and $\mathcal{G}$ on the two subdomains. Finally, the most involved step is combining these subdomain transformations into an orthogonal transformation on the whole domain in Lemma 23.

The following proposition, a nonplanar, orthogonal version of [10, Lemmas 31 and 32], lets us assume $\mathcal{F}$ and $\mathcal{G}$ are quantum-gadget-closed when proving Theorem 12. We defer the proof until after we develop appropriate machinery in section 4 but, intuitively, the first point holds because corresponding $[\mathcal{F}]$ and $[\mathcal{G}]$ grids decompose into corresponding quantum $\mathcal{F}$ and $\mathcal{G}$ grids, and the second point holds because the action of an orthogonal H respects the operations used to construct gadgets.

Lemma 19. *For any signature sets $\mathcal{F}$ and $\mathcal{G}$,*

1. *$\mathcal{F}$ and $\mathcal{G}$ are Holant-indistinguishable iff $[\mathcal{F}]$ and $[\mathcal{G}]$ are Holant-indistinguishable.*
2. *For any orthogonal H, $H\,\mathcal{F} = \mathcal{G}$ iff $H[\mathcal{F}] = [\mathcal{G}]$ (in particular, $H[\mathcal{F}] = [H\,\mathcal{F}]$).*

In the context of #CSP, $\mathcal{EQ}$ enables the gadget-construction of entrywise products of arbitrary signatures. This is a basis for Vandermonde interpolation, a powerful technique in the study of counting problems. In the general Holant setting, we cannot construct the entrywise product of arbitrary signatures, but, for binary signatures whose matrix forms are diagonal, composition is equivalent to entrywise product. We use the following form of Vandermonde interpolation, which follows directly from the similar [30, Lemma 2.3].

Proposition 20. *Let $V \subset \mathbb{R}^q$ be a vector space closed under entrywise product and containing the all-ones vector. For $v \in V$ and $a \in \mathbb{R}$, define the indicator vector $v^a \in \{0,1\}^q$ by $(v^a)_x = 1$ if $v_x = a$ and $(v^a)_x = 0$ otherwise. Then $v^a \in V$ for every $v \in V$ and $a \in \mathbb{R}$.*

Proposition 20 will give us the following matrices, whose diagonals we identify with $\mathbb{R}^q$: For $Z \subset [q]$, define the diagonal matrix (binary signature) $\mathbb{1}_Z \in \{0,1\}^{q\times q}$ by $(\mathbb{1}_Z)_{x,x} = 1$ if $x \in Z$, and $(\mathbb{1}_Z)_{x,x} = 0$ otherwise.

Lemma 21. *Let $\mathcal{F}$ and $\mathcal{G}$, on domain $[q]$, be Holant-indistinguishable, and let $Z \subset [q]$. If $\mathcal{F}$ and $\mathcal{G}$ contain corresponding copies of $\mathbb{1}_Z$, then $\mathcal{F}\,|_Z$ and $\mathcal{G}\,|_Z$ are Holant-indistinguishable.*

Proof. Let Ω be a $\mathcal{F}\,|_Z$-grid. Construct a $\mathcal{F}$-grid Ω' from Ω by replacing every signature $F|_Z \in \mathcal{F}\,|_Z$ with the corresponding $F \in \mathcal{F}$, then replacing every edge with a degree-2 vertex assigned $\mathbb{1}_Z \in \mathcal{F}$. Since $\mathbb{1}_Z$ is 0 when given any input not in Z, any edge assignment sending an edge outside of Z contributes 0 to $\mathrm{Holant}_{\Omega'}(\mathcal{F})$. Furthermore, on inputs from Z, $\mathbb{1}_Z$ acts identically to an edge (as I) in a $\mathrm{Holant}(\mathcal{F}\,|_Z)$ grid. Similar reasoning applies to $\mathcal{G}$ and $\mathcal{G}\,|_Z$, so

$$\mathrm{Holant}_{\Omega}(\mathcal{F}\,|_Z) = \mathrm{Holant}_{\Omega'}(\mathcal{F}) = \mathrm{Holant}_{\Omega'_{\mathcal{F}\to\mathcal{G}}}(\mathcal{G}) = \mathrm{Holant}_{\Omega_{\mathcal{F}\,|_Z\to\mathcal{G}\,|_Z}}(\mathcal{G}\,|_Z). \quad \square$$

The following definition and its applications below borrow a simple but powerful idea of Shao and Cai [46, Section 8.2]: isolating all vertices of an $\mathcal{F} \cup \{F\}$-grid Ω assigned F, the rest of Ω is an $\mathcal{F}$-gadget.

Definition 22 (Subgadget, $\overline{\mathbf{K}}$)**.** *Let* $\mathbf{J}$ *be a gadget. A* subgadget $\mathbf{K} \subset \mathbf{J}$ *induced by a subset* $U \subset V(\mathbf{J})$ *of vertices of* $\mathbf{J}$ *is a gadget composed of the vertices in* U *and* all *of their incident edges: internal edges of* $\mathbf{J}$ *incident to exactly one vertex in* U *become new dangling edges of* $\mathbf{K}$*. For any* $\mathbf{K} \subset \mathbf{J}$*, there is a unique (up to permutation of dangling edges) subgadget* $\overline{\mathbf{K}} \subset \mathbf{J}$*, induced by* $V(\mathbf{J}) \setminus U$*, called the* complement *of* $\mathbf{K}$*, such that, upon reconnecting the new dangling edges of* $\mathbf{K}$ *and* $\overline{\mathbf{K}}$*, we recover* $\mathbf{J}$*.*

We often take $\mathbf{J}$ *to be a signature grid (0-ary gadget)* Ω*, in which case* $\Omega = \langle \mathbf{K}, \overline{\mathbf{K}} \rangle$*.*

We now prove the key inductive lemma. Given a diagonal matrix D capable of separating the domain $[q]$, we apply Lemma 21 to obtain indistinguishability on both subdomains, apply induction to obtain orthogonal transformations on both subdomains, then show how to stitch these subdomain transformations together into a transformation on the whole domain.

Lemma 23. *Let* $\mathcal{F}$ *and* $\mathcal{G}$ *be signature sets on domain* $[q]$*, and suppose Theorem 12 holds for all* $\mathcal{F}'$*,* $\mathcal{G}'$ *on domain smaller than* q*. If* $\mathcal{F}$ *and* $\mathcal{G}$ *are Holant-indistinguishable and contain corresponding copies of a diagonal matrix (binary signature)* $D \notin \operatorname{span}(I)$*, then* $\mathcal{F}$ *and* $\mathcal{G}$ *are ortho-equivalent.*

Proof. By Lemma 19, we may replace $\mathcal{F}$ and $\mathcal{G}$ with $[\mathcal{F}]$ and $[\mathcal{G}]$ to assume $\mathcal{F}$ and $\mathcal{G}$ are quantum-gadget-closed. Since $D \notin \operatorname{span}(I)$, there exist $x, y \in [q]$ such that $D_{x,x} \neq D_{y,y}$, so the sets

$$X = \{z \in [q] : D_{z,z} = D_{x,x}\} \text{ and } Y = [q] \setminus X$$

are a nontrivial partition of $[q]$. Since $\mathcal{F}$ is quantum-gadget-closed, it contains the signature I of the wire. Consider the subalgebra $\langle D, I \rangle_{+,\circ} \subset [\mathcal{F}] = \mathcal{F}$. Since D and I are diagonal, composition $\circ$ is equivalent to entrywise multiplication in $\langle D, I \rangle_{+,\circ}$. Therefore, by Proposition 20 (identifying the matrix diagonals with $\mathbb{R}^q$), we have $\mathbb{1}_X, \mathbb{1}_Y \in \langle D, I \rangle_{+,\circ} \subset \mathcal{F}$. Applying the same interpolation in $\mathcal{G}$, we obtain corresponding copies of $\mathbb{1}_X, \mathbb{1}_Y \in \mathcal{G}$. Now, applying Lemma 21 with $Z := X$, we conclude $\mathcal{F}\,|_X$ and $\mathcal{G}\,|_X$ are Holant-indistinguishable. Furthermore, $|X| < q$, so, by assumption, there is an orthogonal matrix $H_X \in \mathbb{R}^{X \times X}$ satisfying

$$H_X \, \mathcal{F}\,|_X = \mathcal{G}\,|_X. \tag{3}$$

For $b \in [q]$, let $\Delta_b \in \mathbb{R}^q$ be the unary *pinning signature* defined by

$$\Delta_b(x) = \begin{cases} 1 & b = x \\ 0 & b \neq x \end{cases} \text{ for } x \in [q].$$

Call signatures in $\{\Delta_b \mid b \in X\}$ X*-pins*. Let I_Y be the identity operator on $\mathbb{R}^{Y \times Y}$ and define

$$\mathcal{F}' = \mathcal{F} \sqcup \{\Delta_b \mid b \in X\} \text{ and } \mathcal{G}' = \left((H_X^{-1} \oplus I_Y)\, \mathcal{G}\right) \sqcup \{\Delta_b \mid b \in X\}. \tag{4}$$

Claim 24. *$\mathcal{F}'$ and $\mathcal{G}'$ are Holant-indistinguishable.*

Proof. To see this, let Ω be a connected $\mathcal{F}'$-grid. If Ω contains no X-pins then it is an $\mathcal{F}$-grid, and $\Omega_{\mathcal{F}'\to\mathcal{G}'}$ is the corresponding $(H_X^{-1} \oplus I_Y)\,\mathcal{G}$-grid, so by assumption and Corollary 11, $\text{Holant}_\Omega = \text{Holant}_{\Omega_{\mathcal{F}'\to\mathcal{G}'}}$. If Ω contains two adjacent X-pins, then, since Ω is connected, its underlying multigraph consists only of these two vertices. X-pins in $\mathcal{F}'$ correspond to identical X-pins in $\mathcal{G}'$, so again $\text{Holant}_\Omega = \text{Holant}_{\Omega_{\mathcal{F}'\to\mathcal{G}'}}$. Otherwise, Ω contains p pairwise non-adjacent vertices assigned X-pins. Let $\mathbf{K} \subset \Omega$ be a subgadget induced by these p vertices, so $\mathbf{K}$'s signature is $\bigotimes_{i=1}^{p} \Delta_{b_i}$ for some $b_1, \ldots, b_p \in X$. Since $\mathbf{K}$ includes *all* the vertices in Ω assigned X-pins, its complement $\overline{\mathbf{K}}$ is an $\mathcal{F}$-gadget. See Figure 4. Hence the signature F of $\overline{\mathbf{K}}$ is in $\mathcal{F}$, as $\mathcal{F}$ is quantum-gadget-closed. From

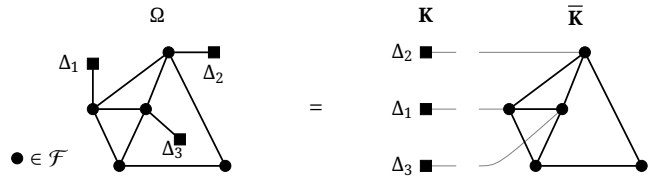


Figure 4: An $\mathcal{F}'$-grid Ω and its pin-induced subgadget $\mathbf{K}$ and complement $\overline{\mathbf{K}}$

$\Omega = \langle \mathbf{K}, \overline{\mathbf{K}} \rangle$, we obtain

$$\text{Holant}_\Omega = \left\langle \bigotimes_{i=1}^{p} \Delta_{b_i}, F \right\rangle = F_{\mathbf{b}},$$

where $\mathbf{b} = b_1 b_2 \ldots b_p$. A similar calculation, using the fact that $\mathcal{G}$ is also quantum-gadget-closed, shows $\text{Holant}_{\Omega_{\mathcal{F}'\to\mathcal{G}'}} = ((H_X^{-1} \oplus I_Y)G)_{\mathbf{b}}$ for $\mathcal{G} \ni G \leftrightsquigarrow F$. Then, since $\mathbf{b} \in X^p$, we have, using (3),

$$\text{Holant}_\Omega = F_{\mathbf{b}} = (F|_X)_{\mathbf{b}} = (H_X^{-1}G|_X)_{\mathbf{b}} = (((H_X^{-1}\oplus I_Y)G)|_X)_{\mathbf{b}} = ((H_X^{-1}\oplus I_Y)G)_{\mathbf{b}} = \text{Holant}_{\Omega_{\mathcal{F}'\to\mathcal{G}'}}$$

(where the fourth equality uses Corollary 54 with $K := G$ and $H := H_X^{-1} \oplus I_Y$). So $\mathcal{F}'$ and $\mathcal{G}'$ are Holant-indistinguishable, completing the proof of Claim 24. □

By Lemma 19, $[\mathcal{F}']$ and $[\mathcal{G}']$ are also Holant-indistinguishable and, by Proposition 13 and Proposition 14, it suffices to show $[\mathcal{F}']$ and $[\mathcal{G}']$ are ortho-equivalent to complete the proof. Note that $[\mathcal{F}']$ and $[\mathcal{G}']$ still contain $\mathbb{1}_Y$ (as $\mathbb{1}_Y$, being 0 on X, is unaffected by the transform $H_X^{-1} \oplus I_Y$), so we may apply Lemma 21 to conclude $\big([\mathcal{F}']\big)\big|_Y$ and $\big([\mathcal{G}']\big)\big|_Y$ are Holant-indistinguishable. Again, since $|Y| < q$, there is an orthogonal $H_Y \in \mathbb{R}^{Y\times Y}$ such that $H_Y\big([\mathcal{F}']\big)\big|_Y = \big([\mathcal{G}']\big)\big|_Y$. Define

$$\mathcal{F}'' = [\mathcal{F}'] \sqcup \{\Delta_b \mid b \in Y\} \text{ and } \mathcal{G}'' = \big((I_X \oplus H_Y^{-1})[\mathcal{G}']\big) \sqcup \{\Delta_b \mid b \in Y\}. \quad (5)$$

By Proposition 13 and Proposition 14, it suffices to show that $\mathcal{F}''$ and $\mathcal{G}''$ are ortho-equivalent. As $[\mathcal{F}']$ and $[\mathcal{G}']$ are, like $\mathcal{F}$ and $\mathcal{G}$, quantum-gadget-closed and Holant-indistinguishable, we repeat the proof of Claim 24, with (5) in place of (4), to show that $\mathcal{F}''$ and $\mathcal{G}''$ are Holant-indistinguishable. Observe that, by definition, $\mathcal{F}''$ and $\mathcal{G}''$ contain corresponding copies of all Y-pins $\{\Delta_b \mid b \in Y\}$. Furthermore, $\mathcal{F}'' \supset [\mathcal{F}'] \supset \mathcal{F}' \supset \{\Delta_b \mid b \in X\}$ and

$$\mathcal{G}'' \supset (I_X \oplus H_Y^{-1})[\mathcal{G}'] \supset (I_X \oplus H_Y^{-1})\,\mathcal{G}' \supset (I_X \oplus H_Y^{-1})\{\Delta_b \mid b \in X\} = \{\Delta_b \mid b \in X\},$$

where the final equality holds because the X-pins are zero on Y, so are unaffected by the transform $(I_X \oplus H_Y^{-1})$. Thus $\mathcal{F}''$ and $\mathcal{G}''$ contain corresponding copies of all pins Δ_b for $b \in [q]$. We claim that this implies that $\mathcal{F}'' = \mathcal{G}''$. To see this, consider any $\mathcal{F}'' \ni F \leftrightsquigarrow G \in \mathcal{G}''$ of common arity n. For any $\mathbf{x} \in [q]^n$, since $\mathcal{F}''$ and $\mathcal{G}''$ are Holant-indistinguishable, we have

$$F_{\mathbf{x}} = \left\langle F, \bigotimes_{i=1}^{n} \Delta_{x_i} \right\rangle = \left\langle G, \bigotimes_{i=1}^{n} \Delta_{x_i} \right\rangle = G_{\mathbf{x}}.$$

Thus $F = G$ for every $\mathcal{F}'' \ni F \leftrightsquigarrow G \in \mathcal{G}''$, so $\mathcal{F}'' = \mathcal{G}''$. □

The final step is to realize the diagonal matrix D in the statement of Lemma 23. To do this, we will apply the duality theory developed in section 4 to prove the following lemma.

Lemma 25. *If $\mathcal{F}$ and $\mathcal{G}$ on domain $[q]$ are Holant-indistinguishable, then there exist $V, U \in O(q)$ such that, after replacing $\mathcal{F}$ by $V\,\mathcal{F}$ and $\mathcal{G}$ by $U\,\mathcal{G}$, there is a nonzero diagonal matrix $D \in \mathbb{R}^{q \times q}$ such that, for every corresponding pair $F \in \mathcal{F}$ and $G \in \mathcal{G}$ with arity $n \geq 2$ and every $m, d \geq 1$ with $m + d = n$, we have*

$$D^{\otimes m} F^{m,d} = G^{m,d} D^{\otimes d}.$$

While the inductive Lemma 23 gives an explicit procedure to construct a transformation between $\mathcal{F}$ and $\mathcal{G}$ given D and the subdomain transformations, the transformations V and U and the matrix D intertwining $\mathcal{F}$ and $\mathcal{G}$ are obtained nonconstructively in the proof of Lemma 25, which applies a black-box invariant-theoretic theorem of Schrijver [43].

Now we combine the inductive Lemma 23 and the nonconstructive Lemma 25 to prove Theorem 12.

Proof of Theorem 12. $(ii) \implies (i)$ is Corollary 11. We show $(i) \implies (ii)$. Let $\mathcal{F}, \mathcal{G}$ be Holant-indistinguishable. We proceed by induction on the domain size q. If $q = 1$ then the only orthogonal matrices in $\mathbb{R}^{q \times q}$ are $\pm I$ and every signature $F \in \mathbb{R}^{[1]^n}$, regardless of arity n, has a single entry, which we denote by $F_0 \in \mathbb{R}$. Let $F \leftrightsquigarrow G$ have even arity n. Contracting $\frac{n}{2}$ pairs of inputs of both F and G, we obtain corresponding signature grids with values F_0 and G_0, respectively, so $F_0 = G_0$, hence $F = G$. For $F \leftrightsquigarrow G$ of odd arity n,

$$F_0^2 = \|F\|^2 = \langle F, F \rangle = \langle G, G \rangle = \|G\|^2 = G_0^2,$$

so $F_0 = \pm G_0$. Let $F \leftrightsquigarrow G$ have odd arity n and $F' \leftrightsquigarrow G'$ have odd arity n', all nonzero. Then $F^{\otimes n'} \otimes (F')^{\otimes n}$ has even arity $2nn'$; contracting its nn' pairs of inputs gives a $\mathcal{F}$-grid with value $F_0^{n'}(F_0')^n$. The corresponding $\mathcal{G}$-grid has value $G_0^{n'}(G_0')^n$. Let $G_0 = (-1)^a F_0$ and $G_0' = (-1)^{a'} F_0'$ for $a, a' \in \{0,1\}$. Then

$$F_0^{n'}(F_0')^n = G_0^{n'}(G_0')^n = (-1)^{an'+a'n} F_0^{n'}(F_0')^n = (-1)^{a+a'} F_0^{n'}(F_0')^n$$

(where in the final equality we used that n, n' are odd), so $a = a'$. Thus there is a common $a \in \{0,1\}$ such that $((-1)^a)^n F = G$ for every n-ary $F \leftrightsquigarrow G$, so $(-I)^a \mathcal{F} = \mathcal{G}$.

Now assume $q > 1$. By Lemma 19, we may assume $\mathcal{F}$ and $\mathcal{G}$ are quantum-gadget-closed. Let $V, U \in O(q)$ be the transformations given by Lemma 25. By Proposition 13, we may replace $\mathcal{F}$ by $V\mathcal{F}$ and $\mathcal{G}$ by $U\mathcal{G}$. Now, by Lemma 25, there is a nonzero diagonal matrix $D \in \mathbb{R}^{q \times q}$ such that, for every corresponding pair $F \in \mathcal{F}$ and $G \in \mathcal{G}$ with arity $n \geq 2$ and every $m, d \geq 1$ with $m + d = n$,

$$D^{\otimes m} F^{m,d} = G^{m,d} D^{\otimes d}. \tag{6}$$

We consider two cases for D: either $D \in \operatorname{span}(I)$ or $D \notin \operatorname{span}(I)$. First, suppose $D \in \operatorname{span}(I)$, so $D = cI$ for $c \neq 0$. Let $F \leftrightsquigarrow G$ be nonzero n-ary signatures with $n \geq 2$ (note that $F = 0 \iff G = 0$ because $\|F\|^2 = \langle F, F\rangle = \langle G, G\rangle = \|G\|^2$). Applying (6) with $m = n - 1$ and $d = 1$ gives

$$c^{n-2} F = G. \tag{7}$$

Then

$$\|F\|^2 = \langle F, F\rangle = \langle G, G\rangle = c^{2(n-2)} \|F\|^2. \tag{8}$$

As $\mathcal{F}$ and $\mathcal{G}$ are quantum-gadget-closed, there are some $\mathcal{F} \ni F \leftrightsquigarrow G \in \mathcal{G}$ with arity $n \geq 3$, so (8) gives $c = \pm 1$. Now applying (7) to any n-ary pair $F \leftrightsquigarrow G$ with $n \geq 2$ gives $c^n F = c^{n-2} F = G$, so $(cI)F = G$, with $cI \in O(q)$. For unary ($n = 1$) F and G, since $\mathcal{F}$ and $\mathcal{G}$ are quantum-gadget-closed, they contain the ternary signatures $F^{\otimes 3}$ and $G^{\otimes 3}$, respectively. So, by the previous reasoning, $(cI)F^{\otimes 3} = G^{\otimes 3}$, or equivalently $(cF)^{\otimes 3} = G^{\otimes 3}$, which implies $cF = G$, as F and G are real-valued. Combining the non-unary and unary cases gives $(cI)\mathcal{F} = \mathcal{G}$.

Otherwise, $D \notin \operatorname{span}(I)$. We will show $\mathcal{F} \cup \{D\}$ and $\mathcal{G} \cup \{D\}$ are Holant-indistinguishable, then apply the inductive Lemma 23. The proof is illustrated in Figure 5. Consider a $\mathcal{F} \cup \{D\}$-grid Ω with at least one vertex assigned D (if Ω has no such vertex then we are done, as $\mathcal{F}$ and $\mathcal{G}$ are Holant-indistinguishable). Let $\mathbf{K} \subset \Omega$ be the subgadget induced by all vertices assigned D. Any connected component of $\mathbf{K}$ is either a cycle or a binary path gadget with signature D^m for some m. The multiplicative factors from corresponding D-cycles in Ω and $\Omega_{\mathcal{F} \cup \{D\} \to \mathcal{G} \cup \{D\}}$ cancel, so, disregarding its cycle components, $\mathbf{K}$ consists of p disconnected path gadgets for some p. By rearranging the dangling edges of $\mathbf{K}$ and $\overline{\mathbf{K}}$, we may assume $\mathbf{K} \in \mathfrak{G}_{\{D\}}(p, p)$ with $M(\mathbf{K}) = \bigotimes_{i=1}^{p} D^{m_i}$ for $m_1, \ldots, m_p \geq 1$, and furthermore that $\overline{\mathbf{K}} \in \mathfrak{G}_{\mathcal{F}}(p, p)$, and that connecting the ith left input and ith right input of $\mathbf{K} \circ \overline{\mathbf{K}}$, for $i \in [p]$, reconstructs Ω (see

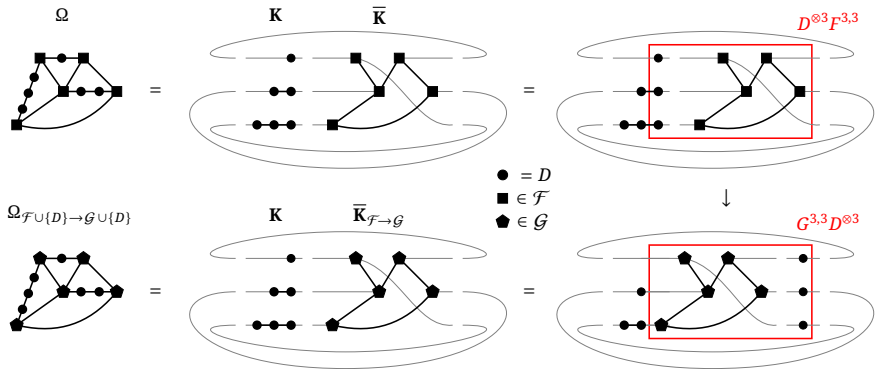


Figure 5: The Holant-value-preserving transformation from Ω to $\Omega_{\mathcal{F}\cup\{D\}\to\mathcal{G}\cup\{D\}}$ in the $D \notin \operatorname{span}(I)$ case. The transition from the bottom right grid to the bottom center grid wraps the three D vertices on the right around to the left along their respective wires.

Figure 5). Since $\overline{\mathbf{K}}$ is an $\mathcal{F}$-gadget and $\mathcal{F}$ is quantum-gadget-closed, $\overline{\mathbf{K}}$ has signature F for some $F \in \mathcal{F}$. Then

$$\operatorname{Holant}_{\Omega} = \operatorname{tr}\left(M(\mathbf{K})M(\overline{\mathbf{K}})\right) = \operatorname{tr}\left(\left(\bigotimes_{i=1}^{p} D^{m_i}\right) F^{p,p}\right). \tag{9}$$

With $F \leftrightsquigarrow G$, we similarly have

$$\operatorname{Holant}_{\Omega_{\mathcal{F}\cup\{D\}\to\mathcal{G}\cup\{D\}}} = \operatorname{tr}\left(\left(\bigotimes_{i=1}^{p} D^{m_i}\right) G^{p,p}\right). \tag{10}$$

Now (9), (6) with $m = d = p$, and (10) give

$$\begin{aligned}
\operatorname{Holant}_{\Omega} &= \operatorname{tr}\left(\left(\bigotimes_{i=1}^{p} D^{m_i}\right) F^{p,p}\right) \\
&= \operatorname{tr}\left(\left(\bigotimes_{i=1}^{p} D^{m_i-1}\right) D^{\otimes p} F^{p,p}\right) \\
&= \operatorname{tr}\left(\left(\bigotimes_{i=1}^{p} D^{m_i-1}\right) G^{p,p} D^{\otimes p}\right) \\
&= \operatorname{tr}\left(D^{\otimes p}\left(\bigotimes_{i=1}^{p} D^{m_i-1}\right) G^{p,p}\right) \\
&= \operatorname{tr}\left(\left(\bigotimes_{i=1}^{p} D^{m_i}\right) G^{p,p}\right) \\
&= \operatorname{Holant}_{\Omega_{\mathcal{F}\cup\{D\}\to\mathcal{G}\cup\{D\}}}.
\end{aligned}$$

Thus $\mathcal{F} \cup \{D\}$ and $\mathcal{G} \cup \{D\}$ are Holant-indistinguishable. This fact, along with the induction hypothesis, lets us apply Lemma 23 to conclude that $\mathcal{F} \cup \{D\}$ and $\mathcal{G} \cup \{D\}$ are ortho-equivalent. Therefore, by Proposition 14, $\mathcal{F}$ and $\mathcal{G}$ are ortho-equivalent. □

4 Quantum Gadget Duality

In this section, we use an invariant-theoretic result of Schrijver [43] to prove a combinatorial-algebraic duality (Theorem 37) showing that quantum $\mathcal{F}$-gadgets exactly capture all tensors invariant under the action of the group of orthogonal transformations stabilizing $\mathcal{F}$. The proof of Theorem 37 entails characterizing quantum $\mathcal{F}$-gadgets as the algebra generated by a few fundamental gadgets under operations respecting the action of $O(q)$. Theorem 37 and its proof are extensions of Regts' [37, Theorem 3] from edge-coloring models to general signature sets $\mathcal{F}$, largely translated into the language of signature matrices. We then use Theorem 37 to prove the key nonconstructive Lemma 25.

We will use the following two notations from invariant theory [37, 43].

Definition 26 ($T(\mathbb{R}^q)^Q$, $\mathrm{Stab}(\mathcal{F})$)**.** *For a subgroup $Q \subset O(q)$, define*

$$T(\mathbb{R}^q)^Q := \{F \in T(\mathbb{R}^q) \mid HF = F \text{ for every } H \in Q\} \subset T(\mathbb{R}^q).$$

Dually, for a signature set $\mathcal{F} \subset T(\mathbb{R}^q)$, define

$$\mathrm{Stab}(\mathcal{F}) := \{H \in O(q) \mid HF = F \text{ for every } F \in \mathcal{F}\} \subset O(q).$$

Definition 27 ($\mathcal{W}$)**.** *Let $\mathcal{W} \subset T(\mathbb{R}^q)$ be the signatures of gadgets with no vertices – that is, the signatures of gadgets consisting of only wires and vertexless loops.*

A gadget consisting of n wires has a $2n$-ary signature in $\mathcal{W}$, which uniquely corresponds, via the order of the gadget's dangling edges, to a partition (or matching) of $[2n]$ into two-element subsets. Since the gadgets defining $\mathcal{W}$ have no vertices, they belong to $\mathfrak{G}_{\mathcal{F}}$ for any $\mathcal{F}$. This view of $\mathcal{W}$ as a set of universal signatures is reinforced by the following classical theorem from representation theory, called the (tensor) *First Fundamental Theorem* for $O(q)$, proved by Weyl [51], and stated in this form by Regts [38, Theorem 4.3]. Define $\langle \mathcal{F} \rangle_+$ as the set of all $\mathbb{R}$-linear combinations of (matching-arity) signatures in $\mathcal{F}$.

Theorem 28 (FFT for $O(q)$)**.** $T(\mathbb{R}^q)^{O(q)} = \langle \mathcal{W} \rangle_+$.

We will not need the $\subset$ inclusion of Theorem 28, although it follows directly from our Theorem 37 below. The $\supset$ inclusion follows from a simple calculation and has the geometric intuition shown in Figure 6.

Say $\mathcal{F} \subset T(\mathbb{R}^q)$ is a *graded subalgebra* of $T(\mathbb{R}^q)$ if $\mathcal{F} \cap (\mathbb{R}^q)^{\otimes n}$ (the signatures of arity n) is a vector space over $\mathbb{R}$ (closed under $\mathbb{R}$-linear combinations) and $\mathcal{F}$ is closed under $\otimes$. Say $\mathcal{F}$ is *contraction-closed* if, for every n-ary $F \in \mathcal{F}$, the $(n-2)$-ary signature resulting from *contracting* any two inputs of F (connecting

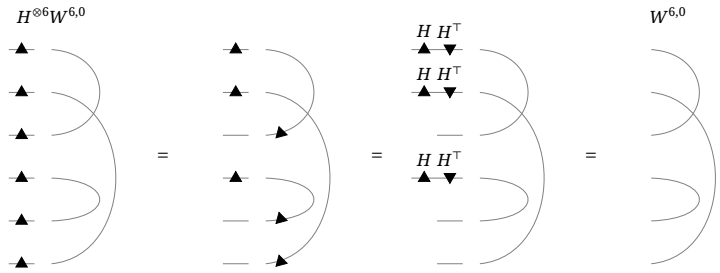


Figure 6: The gadget perspective on the $\supset$ inclusion of Theorem 28, showing $H^{\otimes 6}W^{6,0} = W^{6,0}$ for 6-ary $W \in \mathcal{W}$ and orthogonal H. Each triangle vertex is assigned H. Reassociating indices in the gadget composition expression has the diagrammatic interpretation of moving each lower H along its wire to the top, transposing it in the process (its left and right dangling edges are switched). Then each $HH^\top$ resolves to the identity.

the corresponding dangling edges of the n-ary gadget consisting of a single vertex assigned F) is also in $\mathcal{F}$. The following theorem, due to Schrijver, is the key connection between algebra and combinatorics underlying our proof of Theorem 37 and, eventually, Theorem 12.

Theorem 29 ([43, Corollary 1e])**.** *Let $\mathcal{F} \subset T(\mathbb{R}^q)$. Then there is a subgroup $Q \subset O(q)$ with $\mathcal{F} = T(\mathbb{R}^q)^Q$ if and only if $\mathcal{F}$ is a contraction-closed graded subalgebra of $T(\mathbb{R}^q)$ containing I.*

Schrijver's statement of Theorem 29 requires that $\mathcal{F}$ be "nondegenerate", but it can be seen from the proof that it suffices to assume that $\mathcal{F}$ contains I (cf. [37, Theorem 4]).

Definition 30 $(M(\mathcal{F}), M(T(\mathbb{R}^q)))$**.** *Let $\mathcal{F} \subset T(\mathbb{R}^q)$ be a signature set. Define*

$$M(\mathcal{F}) := \bigcup_{F \in \mathcal{F}} \left(\bigcup_{\substack{m,d \geq 0 \\ m+d = \mathrm{arity}(F)}} F^{m,d} \right)$$

to be the set of all flattenings of all tensors in $\mathcal{F}$. In particular,

$$M(T(\mathbb{R}^q)) = \bigcup_{m,d \geq 0} \mathbb{R}^{q^m \times q^d}.$$

For a subset $P \subset M(T(\mathbb{R}^q))$, define $\langle P \rangle_{+,\circ,\otimes,\top} \subset M(T(\mathbb{R}^q))$ to be the set generated by P under $\circ, \otimes, \top$, and $\mathbb{R}$-linear combinations of matrices with matching dimensions.

Definition 31 (S_σ, S)**.** *For permutation $\sigma \in S_n$, define the $2n$-ary* braid *signature $S_\sigma \in \mathcal{W}$ by $(S_\sigma)_{\mathbf{xy}} = 1$ iff $x_i = y_{\sigma(i)}$ for every $i \in [n]$, and $(S_\sigma)_{\mathbf{xy}} = 0$ otherwise.*

Define the 4-ary 'swap' signature $S := S_{(01)}$. See Figure 7

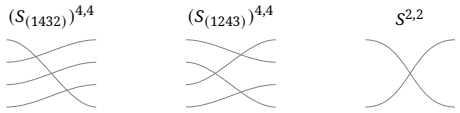


Figure 7: Braid gadgets and their signature matrices.

Every permutation σ is a product of adjacent transpositions, so the flattened braid signature $(S_\sigma)^{n,n} \in \langle I, S^{2,2} \rangle_{\circ,\otimes,\top}$ [55, Lemma 3]. The following is a central object of study in the works of Mančinska, Roberson, Cai and Young [10, 36, 55].

Definition 32. *For $C \subset M(T(\mathbb{R}^q))$, define $C(m,d) := C \cap \mathbb{R}^{q^m \times q^d}$. Say C is a* tensor category with duals *(TCWD) if it satisfies the following properties:*

(i) For every $m, d \geq 0$, $C(m,d)$ is a vector space over $\mathbb{R}$,

(ii) C is closed under $\circ, \otimes, \top$,

(iii) $I \in C(1,1)$,

(iv) $I^{0,2} \in C(0,2)$.

Say C is a symmetric *tensor category with duals if it also satisfies*

(v) $S^{2,2} \in C(2,2)$.

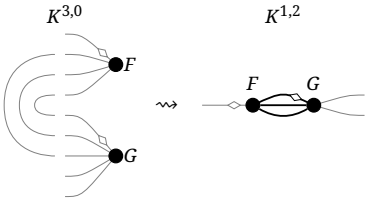
Towards translating Theorem 29 into the language of TCWDs, we next show that a symmetric TCWD is exactly a flattened contraction-closed graded subalgebra of $T(\mathbb{R}^q)$.

Lemma 33. *A subset $\mathcal{F} \subset T(\mathbb{R}^q)$ is a contraction-closed graded subalgebra of $T(\mathbb{R}^q)$ containing I if and only if $M(\mathcal{F})$ is a symmetric tensor category with duals.*

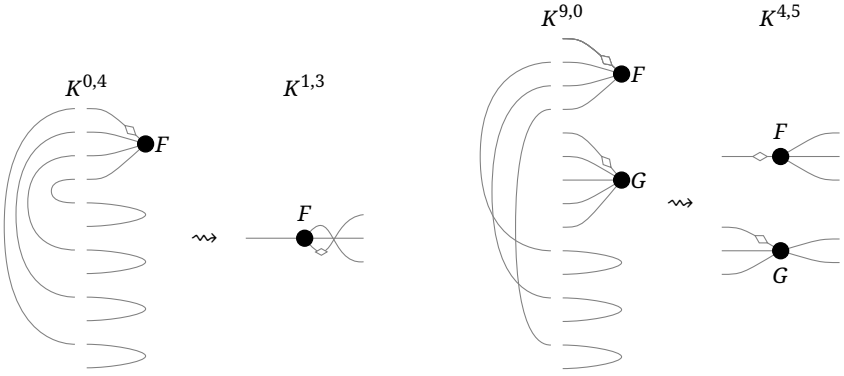
Proof. ($\Longrightarrow$): Let $\mathcal{F}$ be a contraction-closed graded subalgebra of $T(\mathbb{R}^q)$ containing I. As observed by Schrijver [43], for any n-ary $F \in \mathcal{F}$, we can apply any permutation $\sigma \in S_n$ to the inputs of F by constructing $F \otimes I^{\otimes n}$, then contracting the ith input of F with one input of the $\sigma(i)$th copy of I.

First, $I, I^{0,2} \in M(\mathcal{F})$ because $I \in \mathcal{F}$. We can construct $S \in \mathcal{F}$ by contracting inputs to the first and third copies of I and second and fourth copies of I in $I^{\otimes 4} \in \mathcal{F}$, so $S^{2,2} \in M(\mathcal{F})$. Item (i) of Definition 32 holds because $\mathcal{F}$ is graded. Finally, we show that $M(\mathcal{F})$ satisfies item (ii). See Figure 8 (a),(b),(c), respectively.

- $\circ$: Let $F^{m,d}, G^{d,k} \in M(\mathcal{F})$. Construct the signature $K \in \mathcal{F}$ by starting with $F \otimes G \in \mathcal{F}$ and contracting the d lowest-indexed inputs of F (the right/column inputs in $F^{m,d}$) with the d highest-indexed inputs of G, in reverse order. Then $F^{m,d}G^{d,k} = K^{m,k} \in M(\mathcal{F})$.



(a) Constructing $F^{1,3} \circ G^{3,2}$



(c) Constructing $(F^{3,1})^\top$

(b) Constructing $F^{1,3} \otimes G^{3,2}$

Figure 8: Realizing signature matrix $\circ, \otimes, \top$ using contractions, represented by a wire between two dangling edges. Signature inputs are in counterclockwise order, with a diamond indicating the first input.

- $\otimes$: Let $F^{m,d}, G^{m',d'} \in M(\mathcal{F})$. Construct the signature $K \in \mathcal{F}$ by starting with $F \otimes G \in \mathcal{F}$ and reordering the inputs of K as follows: the first m inputs to F, then all inputs to G, then the final d inputs to F. Then $F^{m,d} \otimes G^{m',d'} = K^{m+m',d'+d} \in M(\mathcal{F})$.
- $\top$: Let $F^{m,d} \in M(\mathcal{F})$. Construct the signature $K \in \mathcal{F}$ by reversing the input order of $F \in \mathcal{F}$. Then $(F^{m,d})^\top = K^{d,m} \in M(\mathcal{F})$.

($\Longleftarrow$): Let $M(\mathcal{F})$ be a symmetric TCWD, so $S, I \in \mathcal{F}$, and $\mathcal{F}$ is a graded subspace of $T(\mathbb{R}^q)$ by part (i) of Definition 32. If $F, G \in \mathcal{F}$, then their signature vectors $f, g \in M(\mathcal{F})$, so $f \otimes g \in M(\mathcal{F})$ by closure of $M(\mathcal{F})$ under $\otimes$. Then $F \otimes G \in \mathcal{F}$ because $f \otimes g = (F \otimes G)^{n,0}$ is the signature vector of $F \otimes G$. Thus $\mathcal{F}$ is closed under $\otimes$, so is a graded subalgebra of $T(\mathbb{R}^q)$. Recall that, for any permutation $\sigma \in S_n$, the signature S_σ is constructible from I and S, so $S_\sigma^{n,n} \in M(\mathcal{F})$. Let $F \in \mathcal{F}$ have arity n. To contract inputs i and j of F, define $\sigma \in S_n$ to move inputs i and j to positions 1 and 2, and shift all other inputs down while maintaining their order. Then connect inputs i and j using $I^{0,2}$:

$$(I^{0,2} \otimes I^{\otimes n-2}) \circ S_\sigma^{n,n} \circ f \in M(\mathcal{F}) \tag{11}$$

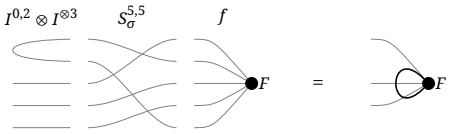


Figure 9: Contracting the 2nd and 5th inputs of F as in (11).

is the signature vector of the ij-contraction of F. See Figure 9. □

Define an action of $O(q)$ on $M(T(\mathbb{R}^q))$ by

$$HF^{m,d} := H^{\otimes m} F^{m,d} (H^\top)^{\otimes d}.$$

Using this action, define the matrix version of Definition 26: for $Q \subset O(q)$ and $\mathcal{F} \subset T(\mathbb{R}^q)$,

$M(T(\mathbb{R}^q))^Q := \{F^{m,d} \in M(T(\mathbb{R}^q)) \mid H^{\otimes m} F^{m,d} (H^\top)^{\otimes d} = F^{m,d} \text{ for every } H \in Q\}$ and
$\mathrm{Stab}(M(\mathcal{F})) := \{H \in O(q) \mid H^{\otimes m} F^{m,d} (H^\top)^{\otimes d} = F^{m,d} \text{ for every } F^{m,d} \in M(\mathcal{F})\}$.

The next proposition shows that the matrix and signature versions of these definitions agree.

Proposition 34.

1. *For any subgroup $Q \subset O(q)$, $M(T(\mathbb{R}^q))^Q$ is a symmetric tensor category with duals.*
2. *For any n-ary signature F and orthogonal H, $(HF)^{m,d} = H^{\otimes m} F^{m,d} (H^\top)^{\otimes d}$ for any $m + d = n$.*
3. *For any $\mathcal{F} \subset T(\mathbb{R}^q)$, $\mathrm{Stab}(\mathcal{F}) = \mathrm{Stab}(M(\mathcal{F}))$.*

Proof. If $F^{m,d}, G^{d,k} \in M(T(\mathbb{R}^q))^Q$ and $H \in Q$ then

$$H^{\otimes m}(F^{m,d}G^{d,k})(H^\top)^{\otimes k} = (H^{\otimes m}F^{m,d}(H^\top)^{\otimes d})(H^{\otimes d}G^{d,k}(H^\top)^{\otimes k}) = F^{m,d}G^{d,k},$$

so $F^{m,d}G^{d,k} \in M(T(\mathbb{R}^q))^Q$. Closure under the other operations $+, \otimes, \top$ follows similarly, so $M(T(\mathbb{R}^q))^Q$ satisfies items (i) and (ii) of Definition 32. Since $I \in \mathcal{W}$, Theorem 28 gives $I \in T(\mathbb{R}^q)^Q$, or $I^{2,0} \in M(T(\mathbb{R}^q))^Q$. Then $I^{0,2} = (I^{2,0})^\top \in M(T(\mathbb{R}^q))^Q$, and $I = I^{1,1} \in M(T(\mathbb{R}^q))^Q$ because $HIH^\top = I$ for $H \in Q \subset O(q)$, giving items (iii) and (iv). Hence $M(T(\mathbb{R}^q))^Q$ is a TCWD.

We now prove part 2. Let $G = HF$. Define the orthogonal block matrix

$$B := \begin{bmatrix} 0 & H^\top \\ H & 0 \end{bmatrix} \tag{12}$$

on $V(F) \sqcup V(G) = [2q]$. Since $H^{\otimes n} f = g$ (so also $(H^\top)^{\otimes n} g = f$), Corollary 55 with $m := n$ and $d := 0$ gives $f \oplus g = (F \oplus G)^{n,0} \in M(T(\mathbb{R}^{2q}))^{\langle B \rangle}$, which, by the previous paragraph, is a TCWD. Now, as in [10, Lemma 3], we use I and $I^{0,2}$ to

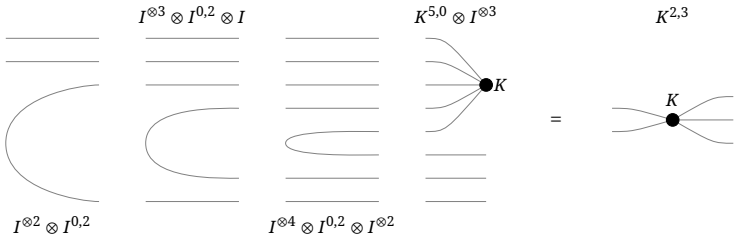


Figure 10: Pivoting three dangling edges of $K = F \oplus G$ from left to right using I and $I^{0,2}$.

'pivot' any number of inputs of $f \oplus g$ from left to right (which preserves the cyclic dangling edge order) to show $(F \oplus G)^{m,d} \in M(T(\mathbb{R}^{2q}))^{\langle B \rangle}$ for any $m + d = n$. See Figure 10. Thus $B^{\otimes m}(F \oplus G)^{m,d} = (F \oplus G)^{m,d} B^{\otimes d}$, so applying the other direction of Corollary 55 gives $H^{\otimes m} F^{m,d} = G^{m,d} H^{\otimes d} = (HF)^{m,d} H^{\otimes d}$, and part 2 follows.

Part 3 follows directly from part 2 and the definitions. Finally, $S \in T(\mathbb{R}^q)^Q$ by Theorem 28 (as $S \in \mathcal{W}$), so $Q \subset \mathrm{Stab}(S) = \mathrm{Stab}(M(\{S\}))$ by part 3. Then $S^{2,2} \in M(T(\mathbb{R}^q))^Q$, giving item (v) of Definition 32 and proving part 1. □

That part 2 of Proposition 34 holds for any matrix H is a well-known fact (see Figure 3). We gave the above proof to mirror the proof of the similar [10, Lemma 6] for quantum orthogonal matrices, and to highlight the role of $I^{0,2}$ in the invariance of a TCWD under edge pivoting.

Since $I, S \in \mathcal{W} \subset [\mathcal{F}]$ for any $\mathcal{F}$, and $\mathfrak{G}_{\mathcal{F}}$ is closed under gadget $\circ, \otimes$, and $\top$, the space $M(\mathfrak{Q}_{\mathcal{F}}) = M([\mathcal{F}])$ is a symmetric TCWD. Specifically, the next proposition, which is similar to [55, Theorem 3], shows that $M(\mathfrak{Q}_{\mathcal{F}})$ is the symmetric TCWD generated by $M(\mathcal{F})$.

Proposition 35. $M(\mathfrak{Q}_{\mathcal{F}}) = \left\langle \left\{ I, I^{0,2}, S^{2,2} \right\} \cup \left\{ f \mid F \in \mathcal{F} \right\} \right\rangle_{+,\circ,\otimes,\top}$.

Proof. $I, S \in \mathcal{W} \subset [\mathcal{F}]$, so $I^{1,1}, I^{0,2}, S^{2,2} \in M(\mathfrak{Q}_{\mathcal{F}})$. The $\supseteq$ direction follows. For the $\subseteq$ direction, it suffices to show that $M(\mathfrak{G}_{\mathcal{F}}) \subseteq \left\langle \left\{ I, I^{0,2}, S^{2,2} \right\} \cup \left\{ f \mid F \in \mathcal{F} \right\} \right\rangle_{\circ,\otimes,\top}$. Let $\mathbf{K} \in \mathfrak{G}_{\mathcal{F}}(m, d)$ be a gadget. By using $I^{2,0} = (I^{0,2})^{\top}$ to pivot dangling edges to the left as in the proof of Proposition 34, we may assume $d = 0$. Suppose $\mathbf{K}$ has p vertices, assigned signatures $F_1, \ldots, F_p$. Break all internal edges of $\mathbf{K}$ and orient the resulting dangling edges to the left to create a gadget with signature

$$\bigotimes_{i=1}^{p} f_i \in \langle f \mid F \in \mathcal{F} \rangle_{\circ,\otimes,\top} . \tag{13}$$

Now we reconstruct $\mathbf{K}$ by reconnecting (contracting) the broken internal edges. Using the formula (11) for arbitrary contraction, we see that $\mathbf{K} \in \left\langle \left\{ I, I^{0,2}, S^{2,2} \right\} \cup \left\{ f \mid F \in \mathcal{F} \right\} \right\rangle_{+,\circ,\otimes,\top}$. □

Proposition 36. $\mathrm{Stab}(\mathcal{F}) = \mathrm{Stab}(M(\mathfrak{Q}_{\mathcal{F}})) = \mathrm{Stab}([\mathcal{F}])$.

Proof. By part 3 of Proposition 34, $\mathrm{Stab}(M(\mathfrak{Q}_{\mathcal{F}})) = \mathrm{Stab}(M([\mathcal{F}])) = \mathrm{Stab}([\mathcal{F}])$. Since $M(\mathfrak{Q}_{\mathcal{F}})$ contains (the vector forms of) $\mathcal{F}$, we have $\mathrm{Stab}(M(\mathfrak{Q}_{\mathcal{F}})) \subseteq \mathrm{Stab}(\mathcal{F})$. By Proposition 35, $M(\mathfrak{Q}_{\mathcal{F}})$ is the symmetric TCWD generated by $\mathcal{F}$ (in particular, is the smallest symmetric TCWD containing $\mathcal{F}$). By part 1 of Proposition 34, $M(T(\mathbb{R}^q))^{\mathrm{Stab}(\mathcal{F})}$ is a symmetric TCWD containing $\mathcal{F}$, so we must have $M(\mathfrak{Q}_{\mathcal{F}}) \subseteq M(T(\mathbb{R}^q))^{\mathrm{Stab}(\mathcal{F})}$, which implies $\mathrm{Stab}(\mathcal{F}) \subseteq \mathrm{Stab}(M(\mathfrak{Q}_{\mathcal{F}}))$. □

Now we come to the main result of this section. We follow Regts' proof for the special case of edge coloring models [38, Theorem 6.8].

Theorem 37. *Let $\mathcal{F}$ be a set of real-valued signatures on domain $[q]$. Then*

$$T(\mathbb{R}^q)^{\mathrm{Stab}(\mathcal{F})} = [\mathcal{F}].$$

Equivalently, by flattening both sides and applying part 3 of Proposition 34, we have, for any m, d,

$$\left(\mathbb{R}^{q^m \times q^d}\right)^{\mathrm{Stab}(\mathcal{F})} = M(\mathfrak{Q}_{\mathcal{F}}(m,d)).$$

Proof. By Proposition 35, $M([\mathcal{F}]) = M(\mathfrak{Q}_{\mathcal{F}})$ is a TCWD and $S \in [\mathcal{F}]$, so $[\mathcal{F}]$ is a contraction-closed graded subalgebra of $T(\mathbb{R}^q)$ containing I by Lemma 33. Thus, by Theorem 29, there is a subgroup $G \subseteq O_q(\mathbb{R})$ such that $[\mathcal{F}] = T(\mathbb{R}^q)^G$. This gives $G \subseteq \mathrm{Stab}([\mathcal{F}])$, so $T(\mathbb{R}^q)^{\mathrm{Stab}([\mathcal{F}])} \subseteq T(\mathbb{R}^q)^G$, and also $T(\mathbb{R}^q)^G = [\mathcal{F}] \subseteq T(\mathbb{R}^q)^{\mathrm{Stab}([\mathcal{F}])}$. Therefore, applying Proposition 36,

$$T(\mathbb{R}^q)^{\mathrm{Stab}(\mathcal{F})} = T(\mathbb{R}^q)^{\mathrm{Stab}([\mathcal{F}])} = T(\mathbb{R}^q)^G = [\mathcal{F}]. \qquad \square$$

A direct consequence is the FFT for $O(q)$. Schrijver [43] gives another proof based on Theorem 29.

Proof of Theorem 28. Putting $\mathcal{F} = \varnothing$ (the empty set) in Theorem 37 gives

$$T(\mathbb{R}^q)^{O(q)} = T(\mathbb{R}^q)^{\mathrm{Stab}(\varnothing)} = [\varnothing] = \langle \mathcal{W} \rangle_+,$$

as $\varnothing$-gadgets must have no vertices. □

Remark 38. *The reasons we translated from Schrijver and Regts's [37, 43] language of contraction-closed graded subalgebras to Mančinska, Roberson, Cai and Young [10, 36, 55]'s language of symmetric TCWDs are twofold. The first is to highlight the similarities between this section's results and the results of the latter group. Enforcing that the TCWDs contain $M(\mathcal{EQ})$ (corresponding to quantum gadgets in the context of* #CSP*), replacing $O(q)$ with S_q (the symmetric group of permutation matrices) or S_q^+ (the* quantum symmetric group*), and dropping the symmetry condition on the TCWDs in the latter case (corresponding to quantum gadget planarity), we find Theorem 29 analogous to the* Tannaka-Krein duality *used by [55] and [10, 36], respectively, to prove results*

analogous to Theorem 37. In fact, the results of this section show that Theorem 29 is equivalent to a well-known classical version of Tannaka-Krein duality [3, Theorem 1.3].

The second reason is that, just as the main result of [10] is a planar, quantum version of the main result of [55], our Theorem 12 should have a planar, quantum version (see Conjecture 52 below). Arbitrary contractions do not respect planarity, but, by simply removing the symmetry condition (i.e. removing $S^{2,2}$ in Proposition 35), we easily enforce planarity in the language of TCWDs.

We now have the tools necessary to prove Lemma 19, which we restate here.

Lemma 19. (Restated) *For any signature sets $\mathcal{F}$ and $\mathcal{G}$,*

1. *$\mathcal{F}$ and $\mathcal{G}$ are Holant-indistinguishable iff $[\mathcal{F}]$ and $[\mathcal{G}]$ are Holant-indistinguishable.*
2. *For any orthogonal H, $H\,\mathcal{F} = \mathcal{G}$ iff $H[\mathcal{F}] = [\mathcal{G}]$ (in particular, $H[\mathcal{F}] = [H\,\mathcal{F}]$).*

Proof. 1: Any $\mathcal{F}$-grid or $\mathcal{G}$-grid is also a $[\mathcal{F}]$-grid or $[\mathcal{G}]$-grid, respectively, giving the ($\Longleftarrow$) direction. For ($\Longrightarrow$), we can express any $[\mathcal{F}]$-grid Ω as a quantum $\mathcal{F}$-grid by, for every vertex v in Ω assigned a signature $F^v \in [\mathcal{F}] \setminus \mathcal{F}$, replacing the subgadget of Ω induced by v by the quantum $\mathcal{F}$-gadget with signature F^v, then linearly expanding to obtain a quantum $\mathcal{F}$-grid. Do the same for $\Omega_{[\mathcal{F}]\to[\mathcal{G}]}$. By assumption, the resulting corresponding quantum $\mathcal{F}$ and $\mathcal{G}$-grids have the same values, so $\text{Holant}_{\Omega} = \text{Holant}_{\Omega_{[\mathcal{F}]\to[\mathcal{G}]}}$.

2: ($\Longleftarrow$) is direct from $\mathcal{F} \subset [\mathcal{F}]$ and $\mathcal{G} \subset [\mathcal{G}]$. For ($\Longrightarrow$), suppose $H\,\mathcal{F} = \mathcal{G}$ for orthogonal H. Consider $\mathcal{F} \oplus \mathcal{G}$ and the block matrix B in (12). As in the proof of Proposition 34, $B \in \text{Stab}(\mathcal{F} \oplus \mathcal{G})$. Then, by Proposition 36, $B \in \text{Stab}([\mathcal{F} \oplus \mathcal{G}])$. Now consider n-ary $[\mathcal{F}] \ni F' \nleftrightsquigarrow G' \in [\mathcal{G}]$, where F' is the signature of $\mathbf{K} \in \mathfrak{Q}_{\mathcal{F}}(n, 0)$, and G' is the signature of $\mathbf{K}_{\mathcal{F}\to\mathcal{G}}$. We may assume no gadget in $\mathbf{K}$ has a component without dangling edges by absorbing the factors from such components (which must be equal for $\mathbf{K}$ and $\mathbf{K}_{\mathcal{F}\to\mathcal{G}}$ by Corollary 11) into the gadget's coefficient, so, by definition of $\oplus$, $M(\mathbf{K}_{\mathcal{F}\to\mathcal{F}\oplus\mathcal{G}})|_{V(\mathcal{F})^n} = M(\mathbf{K}) = f'$ and $M(\mathbf{K}_{\mathcal{F}\to\mathcal{F}\oplus\mathcal{G}})|_{V(\mathcal{G})^n} = M(\mathbf{K}_{\mathcal{F}\to\mathcal{G}}) = g'$ (other blocks are not necessarily 0 if $\mathbf{K}$ is still disconnected). Then, setting $K^{n,0} := M(\mathbf{K}_{\mathcal{F}\to\mathcal{F}\oplus\mathcal{G}}) \in M([\mathcal{F} \oplus \mathcal{G}])$ and considering the equation $K^{n,0} = B^{\otimes n}K^{n,0}$ in block form as in (20), we have

$$\begin{bmatrix} f' \\ * \\ \vdots \\ * \\ g' \end{bmatrix} = \begin{bmatrix} 0 & 0 & \dots & 0 & (H^\top)^{\otimes n} \\ 0 & 0 & \dots & * & 0 \\ \vdots & \vdots & \iddots & \vdots & \vdots \\ 0 & * & \dots & 0 & 0 \\ H^{\otimes n} & 0 & \dots & 0 & 0 \end{bmatrix} \begin{bmatrix} f' \\ * \\ \vdots \\ * \\ g' \end{bmatrix} = \begin{bmatrix} (H^\top)^{\otimes n} g' \\ * \\ \vdots \\ * \\ H^{\otimes n} f' \end{bmatrix}.$$

In particular, $g' = H^{\otimes n} f'$. Therefore $H[\mathcal{F}] = [\mathcal{G}]$. □

Next, we apply the full power of duality (Theorem 37) to prove the nonconstructive part of Lemma 25.

Lemma 39. *If $\mathcal{F}$ and $\mathcal{G}$ are Holant-indistinguishable, then there is a transformation $H \in \mathrm{Stab}(\mathcal{F} \oplus \mathcal{G})$ satisfying $H|_{V(\mathcal{F}),V(\mathcal{G})} \neq 0$ or $H|_{V(\mathcal{G}),V(\mathcal{F})} \neq 0$.*

Proof. Suppose every $H \in \mathrm{Stab}(\mathcal{F} \oplus \mathcal{G})$ has $H|_{V(\mathcal{F}),V(\mathcal{G})} = H|_{V(\mathcal{G}),V(\mathcal{F})} = 0$ (i.e. is block diagonal). Then the block diagonal matrix $A = \begin{bmatrix} I & 0 \\ 0 & 2I \end{bmatrix}$ satisfies $HA = AH$, hence $HAH^T = A$, for every $H \in \mathrm{Stab}(\mathcal{F} \oplus \mathcal{G})$. Thus, by Theorem 37 with $m = d = 1$, A is realizable as the signature matrix of a binary quantum gadget – that is, there exist binary $(\mathcal{F} \oplus \mathcal{G})$-gadgets $\mathbf{K}^1, \dots \mathbf{K}^p$ and $c_1, \dots, c_p \in \mathbb{R}$ such that

$$A = \sum_{i=1}^{p} c_i M(\mathbf{K}^i). \tag{14}$$

Any connected component of a gadget $\mathbf{K}^i$ disconnected from the component(s) of $\mathbf{K}^i$ containing the two dangling edges contributes only an overall multiplicative factor; by absorbing this factor into c_i, we may assume each $\mathbf{K}^i$ has no components without a dangling edge. By definition of $\oplus$, inputting an $x \in V(\mathcal{F})$ along a dangling edge of $\mathbf{K}^i$ forces any edge assignment with nonzero value to assign an element of $V(\mathcal{F})$ to every edge in that dangling edge's connected component. So, for any $x, y \in V(\mathcal{F})$, we have $M(\mathbf{K}^i)_{x,y} = M(\mathbf{K}^i_{(\mathcal{F} \oplus \mathcal{G}) \to \mathcal{F}})_{x,y}$. Similar reasoning applies to $\mathcal{G}$, so the $V(\mathcal{F}), V(\mathcal{F})$ and $V(\mathcal{G}), V(\mathcal{G})$ blocks of (14) are

$$I = \sum_{i=1}^{p} c_i M(\mathbf{K}^i_{(\mathcal{F} \oplus \mathcal{G}) \to \mathcal{F}}) \text{ and } 2I = \sum_{i=1}^{p} c_i M(\mathbf{K}^i_{(\mathcal{F} \oplus \mathcal{G}) \to \mathcal{G}}), \tag{15}$$

respectively. Let Ω^i be the $\mathcal{F}$-grid resulting from connecting the two dangling edges of $\mathbf{K}^i_{(\mathcal{F} \oplus \mathcal{G}) \to \mathcal{F}}$. Then taking the trace of the equations in (15) gives

$$\sum_{i=1}^{p} c_i \operatorname{Holant}_{\Omega^i}(\mathcal{F}) = q \neq 2q = \sum_{i=1}^{p} c_i \operatorname{Holant}_{\Omega^i_{\mathcal{F} \to \mathcal{G}}}(\mathcal{G}),$$

so there is some i for which $\operatorname{Holant}_{\Omega^i}(\mathcal{F}) \neq \operatorname{Holant}_{\Omega^i_{\mathcal{F} \to \mathcal{G}}}(\mathcal{G})$. □

Finally, we apply some postprocessing to Lemma 39 to prove Lemma 25, which we restate here.

Lemma 25. (Restated) *If $\mathcal{F}$ and $\mathcal{G}$ on domain $[q]$ are Holant-indistinguishable, then there exist $V, U \in O(q)$ such that, after replacing $\mathcal{F}$ by $V\mathcal{F}$ and $\mathcal{G}$ by $U\mathcal{G}$, there is a nonzero diagonal matrix $D \in \mathbb{R}^{q \times q}$ such that, for every corresponding pair $F \in \mathcal{F}$ and $G \in \mathcal{G}$ with arity $n \geq 2$ and every $m, d \geq 1$ with $m + d = n$, we have*

$$D^{\otimes m} F^{m,d} = G^{m,d} D^{\otimes d}.$$

Proof. By Lemma 39, there is an $H \in \mathrm{Stab}(\mathcal{F} \oplus \mathcal{G})$ with either $H|_{\mathcal{F},\mathcal{G}} \neq 0$ or $H|_{\mathcal{G},\mathcal{F}} \neq 0$. Assume WLOG that $H|_{\mathcal{G},\mathcal{F}} \neq 0$. Let $H|_{\mathcal{G},\mathcal{F}} = U^\top D V$ be the singular value decomposition of $H|_{\mathcal{G},\mathcal{F}}$, with U, V orthogonal and $D \neq 0$

diagonal, all real. Replace $\mathcal{F}$ with $V\,\mathcal{F}$ and $\mathcal{G}$ with $U\,\mathcal{G}$. This has the effect of replacing $\mathcal{F} \oplus \mathcal{G}$ by $(V\,\mathcal{F}) \oplus (U\,\mathcal{G}) = (V \oplus U)(\mathcal{F} \oplus \mathcal{G})$ (by (19)), which replaces $\mathrm{Stab}(\mathcal{F} \oplus \mathcal{G})$ by $(V \oplus U) \circ \mathrm{Stab}(\mathcal{F} \oplus \mathcal{G}) \circ (V \oplus U)^{-1} = (V \oplus U) \circ \mathrm{Stab}(\mathcal{F} \oplus \mathcal{G}) \circ (V \oplus U)^{\top}$. In particular, H is replaced by

$$\begin{bmatrix} V & 0 \\ 0 & U \end{bmatrix} \begin{bmatrix} H|_{\mathcal{F},\mathcal{F}} & H|_{\mathcal{F},\mathcal{G}} \\ U^{\top}DV & H|_{\mathcal{G},\mathcal{G}} \end{bmatrix} \begin{bmatrix} V^{\top} & 0 \\ 0 & U^{\top} \end{bmatrix} = \begin{bmatrix} VH|_{\mathcal{F},\mathcal{F}}V^{\top} & VH|_{\mathcal{F},\mathcal{G}}U^{\top} \\ U(U^{\top}DV)V^{\top} & UH|_{\mathcal{G},\mathcal{G}}U^{\top} \end{bmatrix} = \begin{bmatrix} * & * \\ D & * \end{bmatrix}.$$

To summarize, after transforming $\mathcal{F}$ by V and $\mathcal{G}$ by U, we have $H = \begin{bmatrix} * & * \\ D & * \end{bmatrix} \in \mathrm{Stab}(\mathcal{F} \oplus \mathcal{G})$ for nonzero diagonal D. Let $F \in \mathcal{F}$ and $G \in \mathcal{G}$ be corresponding signatures of arity $n \geq 2$. For any $m, d \geq 1$ with $m + d = n$, part 3 of Proposition 34 gives

$$H^{\otimes m}(F \oplus G)^{m,d} = (F \oplus G)^{m,d}H^{\otimes d}. \tag{16}$$

Now, by Proposition 53 with $K := F \oplus G$ and (2), we can write (16) as the block matrix equation

$$\begin{bmatrix} * & * & \dots & * \\ \vdots & \vdots & \iddots & \vdots \\ * & * & \dots & * \\ D^{\otimes m} & * & \dots & * \end{bmatrix} \begin{bmatrix} F^{m,d} & 0 & \dots & 0 & 0 \\ 0 & 0 & \dots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \dots & 0 & 0 \\ 0 & 0 & \dots & 0 & G^{m,d} \end{bmatrix} = \begin{bmatrix} F^{m,d} & 0 & \dots & 0 & 0 \\ 0 & 0 & \dots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \dots & 0 & 0 \\ 0 & 0 & \dots & 0 & G^{m,d} \end{bmatrix} \begin{bmatrix} * & * & \dots & * \\ \vdots & \vdots & \iddots & \vdots \\ * & * & \dots & * \\ D^{\otimes d} & * & \dots & * \end{bmatrix} \tag{17}$$

(note the importance of $m, d \geq 1$ so that $F^{m,d}$ and $G^{m,d}$ are not in the same block-column or block-row of the block matrices in (17)). The bottom left block of (17) is $D^{\otimes m}F^{m,d} = G^{m,d}D^{\otimes d}$ as desired. □

5 Consequences of Theorem 12

In this section, we exploit the expressiveness of the Holant framework to show that Theorem 12 encompasses a variety of existing results, and derive a few novel consequences.

5.1 Counting CSP and graph homomorphisms

Recall the set $\mathcal{EQ}$ of equality signatures from Section 2.1. For signature set $\mathcal{F}$, define the *counting constraint satisfaction problem* $\#\mathrm{CSP}(\mathcal{F})$ with *constraint function* set $\mathcal{F}$ to be the problem $\mathrm{Holant}(\mathcal{F} \mid \mathcal{EQ})$. Vertices assigned signatures in $\mathcal{F}$ and $\mathcal{EQ}$ are *constraints* and *variables*, respectively, and a $(\mathcal{F} \mid \mathcal{EQ})$-grid Ω is a constraint-variable incidence graph, where a variable appears in all of its incident constraints (the equality signatures force each variable to take the same value in all constraints in which it appears). Then $\mathrm{Holant}_{\Omega}(\mathcal{F} \mid \mathcal{EQ})$ is the sum over all variable assignments of the product of the constraint evaluations.

Like Holant, #CSP is a well-studied problem in counting complexity, with dichotomy theorems classifying #CSP($\mathcal{F}$) as either tractable or #P-hard proved for increasingly broad classes of constraint function sets [5, 6, 15, 27].

By inserting a dummy degree-2 constraint vertex assigned $(=_2) \in \mathcal{EQ}$ between adjacent variable vertices and combining adjacent constraint vertices assigned $=_a$ and $=_b$ into a single constraint vertex assigned $=_{a+b-2}$, we see that Holant($\mathcal{F} \cup \mathcal{EQ}$) is equivalent to Holant($\mathcal{F} \mid \mathcal{EQ}$). See Figure 11. Say $\mathcal{F}$ and

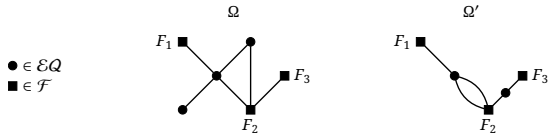


Figure 11: A Holant($\mathcal{F} \cup \mathcal{EQ}$) grid Ω and the equivalent Holant($\mathcal{F} \mid \mathcal{EQ}$) grid Ω'.

$\mathcal{G}$ are *isomorphic* if there exists a permutation matrix H satisfying $H\mathcal{F} = \mathcal{G}$ (in other words, $\mathcal{F}$ and $\mathcal{G}$ are the same up to relabeling of their domains). Say that $\mathcal{F}$ and $\mathcal{G}$ are #CSP-indistinguishable if $\mathcal{F} \cup \mathcal{EQ}$ and $\mathcal{G} \cup \mathcal{EQ}$ are Holant-indistinguishable (in other words, every #CSP instance has the same value whether we use constraint functions from $\mathcal{F}$ or from $\mathcal{G}$). If $\mathcal{F}$ and $\mathcal{G}$ are isomorphic, then they are #CSP-indistinguishable. Applying Theorem 12 to $\mathcal{F} \cup \mathcal{EQ}$ and $\mathcal{G} \cup \mathcal{EQ}$, we obtain the converse of this statement, the main result of Young [55] for real-valued constraint functions.

Corollary 40. *Let $\mathcal{F}$ and $\mathcal{G}$ be sets of real-valued constraint functions. Then $\mathcal{F}$ and $\mathcal{G}$ are isomorphic if and only if $\mathcal{F}$ and $\mathcal{G}$ are* #CSP*-indistinguishable.*

Proof. We only need ($\Longleftarrow$). If $\mathcal{F}$ and $\mathcal{G}$ are #CSP-indistinguishable, then $\mathcal{F} \cup \mathcal{EQ}$ and $\mathcal{G} \cup \mathcal{EQ}$ are Holant-indistinguishable, so, by Theorem 12, there is an orthogonal H such that $H\mathcal{F} = \mathcal{G}$ and $H\mathcal{EQ} = \mathcal{EQ}$. Using a standard Vandermonde interpolation argument, Xia [53] shows that any matrix H satisfies $H\mathcal{EQ} = \mathcal{EQ}$ if and only H is a permutation matrix. Therefore $\mathcal{F}$ and $\mathcal{G}$ are isomorphic. □

Let $\mathcal{EQ}_2 \subset \mathcal{EQ}$ be the set of equality signatures of even arity and $\#\mathrm{CSP}^2(\mathcal{F}) :=$ Holant($\mathcal{F} \mid \mathcal{EQ}_2$) be #CSP($\mathcal{F}$) restricted to instances in which every variable appears an even number of times [14, 31]. We now obtain a sharpening of Corollary 40. A *signed permutation matrix* is an orthogonal matrix with entries in $\{0, \pm 1\}$ and exactly one nonzero entry in each row and column.

Corollary 41. *Let $\mathcal{F}$ and $\mathcal{G}$ be sets of real-valued constraint functions. Then there is a signed permutation matrix P satisfying $\mathcal{G} = P\mathcal{F}$ if and only if $\mathcal{F}$ and $\mathcal{G}$ are* #CSP^2*-indistinguishable.*

Proof. As above, $\#\mathrm{CSP}^2(\mathcal{F}) \equiv$ Holant($\mathcal{F} \mid \mathcal{EQ}_2$) is equivalent to Holant($\mathcal{F} \cup \mathcal{EQ}_2$)

(critically, if $=_a \in \mathcal{EQ}_2$ and $=_b \in \mathcal{EQ}_2$, then $=_{a+b-2} \in \mathcal{EQ}_2$). Schrijver [43] shows[1] that any invertible matrix H satisfies $H\,\mathcal{EQ}_2 = \mathcal{EQ}_2$ if and only if H is a signed permutation matrix. Now, as in the proof of Corollary 40, the result follows from applying Theorem 12 to $\mathcal{F} \cup \mathcal{EQ}_2$ and $\mathcal{G} \cup \mathcal{EQ}_2$. □

As discussed around Figure 1, Holant$(A_X \mid \mathcal{EQ}) \equiv \#\mathrm{CSP}(A_X)$ counts the number of homomorphisms to graph X from multigraphs K. Therefore it follows from Corollary 40 that two graphs X and Y are isomorphic if and only if they admit the same number of homomorphisms from every multigraph K. Furthermore, it follows directly from the definition that the number of homomorphisms from a multigraph K to graph X equals the number of homomorphisms from K' to X, where K' is the graph constructed from K by removing all loops and duplicate edges. Hence Corollary 40 is a generalization of the classical theorem of Lovász [34] that two graphs are isomorphic if and only if they admit the same number of homomorphisms from every graph. Similarly, Corollary 41 characterizes homomorphism indistinguishability from multigraphs in which all vertices have even degree. However, this gives no new information about homomorphisms from graphs, as duplicating all edges of any graph K' yields a multigraph with only even-degree vertices. Nevertheless, by a slightly more involved application of Corollary 41, we obtain the following sharpening of Lovász's theorem (which also follows from a *connector* argument in the style of [26]).

Corollary 42. *Graphs X and Y are isomorphic if and only if they admit the same number of homomorphisms from those graphs in which all vertices have even degree.*

Proof. We only need ($\Longleftarrow$). Consider Holant$(A_X^3 \mid \mathcal{EQ}_2) \equiv \#\mathrm{CSP}^2(A_X^3)$ and Holant$(A_Y^3 \mid \mathcal{EQ}_2) \equiv \#\mathrm{CSP}^2(A_Y^3)$ (where A_X^3 is the third power of matrix A_X). Let Ω be a Holant$(A_X^3 \mid \mathcal{EQ}_2)$ grid, and K_Ω be the multigraph underlying Ω in the sense of Figure 1 (ignoring the degree-2 vertices assigned A_X^3). Now replace every vertex in Ω assigned A_X^3 with a path of five vertices assigned A_X, $=_2$, A_X, $=_2$, A_X (a gadget with signature matrix $A_X I A_X I A_X = A_X^3$). The Holant value of the resulting Holant$(A_X \mid \mathcal{EQ}_2)$ grid Ω' equals the Holant value of Ω, and, following Figure 1, equals the number of homomorphisms to X from a graph (not multigraph) $K_{\Omega'}$ in which all vertices have even degree, as $K_{\Omega'}$ is constructed from K_Ω – a multigraph in which all vertices have even degree – by subdividing every edge twice to eliminate all loops and parallel edges. Applying the assumption and similar reasoning to the Y side, we conclude that A_X^3 and A_Y^3 are $\#\mathrm{CSP}^2$-indistinguishable. Thus Corollary 41 gives a signed permutation matrix P' satisfying $P' A_X^3 (P')^\top = A_Y^3$. Since A_X^3 and A_Y^3 have nonnegative entries and the action of P' only rearranges and possibly negates

[1] The First Fundamental Theorem for $S_q^\pm \subset O(q)$ (the group of signed permutation matrices) states that $T(\mathbb{R}^q)^{S_q^\pm} = [\mathcal{EQ}_2]$ (cf. Theorem 28, and recall that $\langle \mathcal{W} \rangle_+ = [\varnothing]$). It follows as in the proof of Theorem 37 that $\mathrm{Stab}(\mathcal{EQ}_2) = S_q^\pm$. The fact that $\mathrm{Stab}(\mathcal{EQ}) = S_q \subset O(q)$ (the group of permutation matrices) similarly follows from the First Fundamental Theorem for S_q, which states that $T(\mathbb{R}^q)^{S_q} = [\mathcal{EQ}]$.

the entries of A_X^3, there must in fact be no negation, so $PA_X^3P^\top = A_Y^3$, where P is the permutation matrix created by flipping every -1 entry of P' to 1. Since A_X^3 and A_Y^3 are diagonalizable with real eigenvalues (in particular, the eigenvalues have unique cube roots), it follows that $PA_XP^\top = A_Y$, so X and Y are isomorphic. □

5.2 Simultaneous matrix similarity

Let $\mathcal{F}$ and $\mathcal{G}$ be sets of binary signatures, thought of as matrices. Any connected $\mathcal{F}$-grid Ω is a cycle. Breaking an edge of the cycle, we obtain a binary path gadget with signature matrix $\prod_{i=1}^{c} F_i$, where, depending on its orientation, each $F_i \in \mathcal{F}$ or $F_i^\top \in \mathcal{F}$. Connecting the path's two dangling edges, we reform Ω, which thus has Holant value $\operatorname{tr}(\prod_{i=1}^{c} F_i)$. Let $\Gamma_{\mathcal{F}}$ be the set of all finite products of matrices in $\mathcal{F}$ and $\mathcal{F}^\top := \{F^\top \mid F \in \mathcal{F}\}$. Define $\Gamma_{\mathcal{G}}$ similarly and, for a word $w \in \Gamma_{\mathcal{F}}$, construct $w_{\mathcal{F}\to\mathcal{G}} \in \Gamma_{\mathcal{G}}$ by replacing every character F or $F^\top$ in w by the corresponding G or $G^\top$, respectively. For orthogonal H, we have $H\mathcal{F} = \mathcal{G} \iff HF^{1,1} = G^{1,1}H$ for every $F \rightsquigarrow G$ (by part 2 of Proposition 34), so, in this setting, Theorem 12 is equivalent to the following real-valued case of a classical theorem from representation theory, due to Specht [47] and Wiegmann [52]. Grohe, Rattan, and Seppelt [30] also give a combinatorial proof.

Corollary 43. *Let $\mathcal{F}, \mathcal{G} \subset \mathbb{R}^{q\times q}$ be similar. Then there is an $H \in O(q)$ such that $HF = GH$ for every corresponding $F \in \mathcal{F}$ and $G \in \mathcal{G}$ if and only if $\operatorname{tr}(w) = \operatorname{tr}(w_{\mathcal{F}\to\mathcal{G}})$ for every $w \in \Gamma_{\mathcal{F}}$.*

Suppose $\mathcal{F} = \{A_X\}$ and $\mathcal{G} = \{A_Y\}$ for graphs X and Y. Transform an A_X-grid Ω to a $(A_X \mid \mathcal{EQ})$-grid Ω' by inserting a dummy degree-2 vertex assigned $I = (=_2) \in \mathcal{EQ}$ between every consecutive pair of vertices in the cycle. Recall from Section 2.1 that $\operatorname{Holant}_{\Omega'}(A_X \mid \mathcal{EQ})$ counts the number of homomorphisms from graph K to X, where K is the graph obtained from Ω' by ignoring the vertices assigned A_X. Here K is a cycle, so we have the following well-known result, an alternate formulation of this case of Corollary 43.

Corollary 44. *Let X and Y be graphs. Then there is an orthogonal matrix H satisfying $HA_X = A_YH$ if and only if X and Y admit the same number of homomorphisms from all cycles.*

A matrix H is *pseudo-stochastic* if all of its rows and columns sum to 1. Dell, Grohe, and Rattan [22] proved that graphs X and Y admit the same number of homomorphisms from all paths if and only if there is a pseudo-stochastic matrix H such that $HA_X = A_YH$. Using Theorem 12, we combine this result with Corollary 44, which also reproduces a combinatorial explanation for the connection between pseudo-stochastic matrices and homomorphisms from paths [30].

Corollary 45. *Let X and Y be graphs. Then there is a pseudo-stochastic orthogonal matrix H satisfying $HA_X = A_YH$ if and only if X and Y admit the same number of homomorphisms from all cycles and paths.*

Proof. Consider Holant($A_X \cup \{=_1\}$). Any $A_X \cup \{=_1\}$-grid is a disjoint union of cycles composed of signatures assigned A_X and paths with degree-2 internal vertices assigned A_X and degree-1 endpoints assigned $=_1 \in \mathcal{EQ}$. As discussed before Corollary 44, every cycle A_X-grid Ω has the same Holant value as $\Omega_{A_X \to A_Y}$ if and only if X and Y admit the same number of homomorphisms from every cycle. Similarly inserting dummy vertices assigned $=_2$ between every pair of A_X vertices in a path component, we produce a $(A_X \mid \mathcal{EQ})$-grid whose Holant value equals the number of homomorphisms to X from the underlying path. Thus X and Y admit the same number of homomorphisms from all cycles and all paths if and only if $A_X \cup \{=_1\}$ and $A_Y \cup \{=_1\}$ are Holant-indistinguishable. By Theorem 12, this is equivalent to the existence of an orthogonal H satisfying $HA_X = A_Y H$ and $H(=_1) = (=_1)$. The vector form of $=_1$ is the all-ones vector, so $H(=_1) = (=_1)$ if and only if the rows of H sum to 1 and (since $H(=_1) = (=_1) \iff H^\top(=_1) = (=_1)$) the columns of H sum to 1. □

5.3 Odeco signature sets

In this section, we derive a consequence of Theorem 12 that is not a counting indistinguishability theorem, but a combinatorial characterization of signatures that are simultaneously 'diagonalizable'.

Definition 46 ($\mathcal{GEQ}$, odeco)**.** *Define the set of* general equalities *(or* weighted equalities*) on domain* $[q]$ *as* $\mathcal{GEQ} = \{=_n^{\mathbf{a}} \mid n \geq 0, \mathbf{a} \in \mathbb{R}^{[q]}\}$*, where* $=_n^{\mathbf{a}}$ *is the symmetric* n*-ary signature defined by*

$$(=_n^{\mathbf{a}})_{\mathbf{x}} = \begin{cases} a_i & x_1 = \ldots = x_n = i \\ 0 & \text{otherwise.} \end{cases}$$

for $\mathbf{x} \in [q]^n$*. A set* $\mathcal{F}$ *of symmetric signatures is* orthogonally decomposable*, or* odeco*, if it is ortho-equivalent to a general equality set – that is, there exists an* $H \in O(q)$ *such that* $H\,\mathcal{F} \subset \mathcal{GEQ}$*.*

The term "odeco" was coined by Robeva [41] to refer to individual symmetric tensors (signatures) which are ortho-equivalent to a general equality. A binary $\mathcal{GEQ}$ signature has a diagonal signature matrix, so the spectral theorem states that every (real) symmetric binary signature is odeco (recall part 2 of Proposition 34). Any nonzero edge assignment for a connected $\mathcal{GEQ}$-gadget $\mathbf{K}$ must assign all edges, including dangling edges, the same domain element, so, if $\mathbf{K}$ has arity n and is composed of vertices assigned signatures with weights $\mathbf{a}^1, \ldots, \mathbf{a}^p$, then $\mathbf{K}$ has signature $=_n^{\mathbf{a}^1 \bullet \cdots \bullet \mathbf{a}^p} \in \mathcal{GEQ}$, where $\bullet$ denotes entrywise product. In particular, if $\mathbf{K}$ is a $\mathcal{GEQ}$-grid Ω, then $\text{Holant}_\Omega = \sum_{i=1}^q (\mathbf{a}^1 \bullet \ldots \bullet \mathbf{a}^p)_i$. Thus, if $\mathcal{F}$ is odeco, then the computational problem Holant($\mathcal{F}$) is polynomial-time tractable via orthogonal Holographic transformation.

Definition 47 ($*$)**.** *For symmetric signatures* $F_1, F_2 \in \mathcal{F}$ *of arity* n_1 *and* n_2*, respectively, construct the* $(n_1 + n_2 - 2)$*-ary signature* $F_1 * F_2 \in [\mathcal{F}]$ *from* $F_1 \otimes F_2$ *by contracting an input of* F_1 *and an input of* F_2*. See Figure 12.*

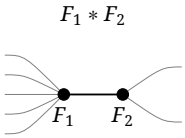


Figure 12: Illustrating (the gadget with signature) $F_1 * F_2$ for 6-ary F_1 and 3-ary F_2.

Since F_1 and F_2 are symmetric, $F_1 * F_2$ doesn't depend on which inputs we connect. For $\mathbf{x} \in [q]^{n_1-1}$ and $\mathbf{y} \in [q]^{n_2-1}$, we have (with vectors viewed as input lists)

$$(F_1 * F_2)(\mathbf{x}, \mathbf{y}) = \sum_{z \in [q]} F_1(\mathbf{x}, z) F_2(\mathbf{y}, z).$$

Proposition 48. *For any $H \in O(q)$ and signature $F \in (\mathbb{R}^q)^{\otimes n}$ of arity $n \geq 3$, we have $HF \in \mathcal{GEQ}$ if and only if $H(F * F) \in \mathcal{GEQ}$.*

Proof. ($\Longrightarrow$): If $HF = E$ for $H \in O(q)$ and $E \in \mathcal{GEQ}$, then by part 2 of Lemma 19, $H(F * F) = (HF) * (HF) = E * E \in \mathcal{GEQ}$.

($\Longleftarrow$): Let $\mathcal{GEQ} \ni E = H(F * F) = (HF) * (HF)$ for $H \in O(q)$. Suppose toward contradiction that $HF \notin \mathcal{GEQ}$, so there is a $\mathbf{x} \in [q]^n$ such that $(HF)(\mathbf{x}) \neq 0$ but $\exists i, j$ such that $x_i \neq x_j$. Assume WLOG that $i, j \neq n$, as $n \geq 3$. Construct $\mathbf{x}' \in [q]^{n-1}$ by deleting the nth (last) entry of $\mathbf{x}$. Then

$$E(\mathbf{x}', \mathbf{x}') = ((HF) * (HF))(\mathbf{x}', \mathbf{x}') = \sum_{z \in [q]} (HF)(\mathbf{x}', z)^2 \geq (HF)(\mathbf{x})^2 > 0,$$

contradicting $E \in \mathcal{GEQ}$. Thus $HF \in \mathcal{GEQ}$. □

Theorem 49. *Let $\mathcal{F}$ be a set of real-valued symmetric signatures (tensors). The following are equivalent.*

(i) $\mathcal{F}$ is odeco.

(ii) Every connected $\mathcal{F}$-gadget has a symmetric signature.

(iii) For every $F_1, F_2 \in \mathcal{F}$, the signature $F_1 * F_2$ is symmetric.

Robeva [41] conjectured the equivalence of items (i) and (iii) when $\mathcal{F}$ contains a single signature. Boralevi, Draisma, Horobeţ, and Robeva [4] confirmed this conjecture using techniques from algebraic geometry. We use Theorem 12 to give a combinatorial proof, generalized to arbitrary symmetric signature sets.

Remark 50. *If $\mathcal{F}$ is a set of symmetric binary signatures, then $F_1 * F_2 = F_1 \circ F_2 = F_1 F_2$ (a matrix product) for $F_1, F_2 \in \mathcal{F}$. In general, symmetric matrices commute if and only if their product is symmetric (as if $F_1 F_2$ is symmetric then $F_1 F_2 = (F_1 F_2)^\top = F_2^\top F_1^\top = F_2 F_1$ and if F_1 and F_2 commute then $(F_1 F_2)^\top =$*

$F_2^\top F_1^\top = F_2F_1 = F_1F_2$*). Therefore Theorem 49 encompasses the extension of the spectral theorem which states that commuting symmetric matrices are simultaneously diagonalizable. We use this fact in the proof below.*

Proof of Theorem 49. (i) $\implies$ (ii),(iii): Suppose $H\mathcal{F} \subset \mathcal{GEQ}$ for some $H \in O(q)$. Let $K \in [\mathcal{F}]$ be the signature of a connected $\mathcal{F}$-gadget (e.g. $K = F_1 * F_2$). By part 2 of Lemma 19, $HK = J$, where J is the signature of a connected $\mathcal{GEQ}$-gadget. Then $J \in \mathcal{GEQ}$, so J, and hence $K = H^{-1}J$, are symmetric.

(ii) $\implies$ (i): First, replace every non-unary odd-arity $F \in \mathcal{F}$ by $F * F$ (every unary signature is in $\mathcal{GEQ}$, so simply remove all unaries from $\mathcal{F}$). This does not change the fact that $\mathcal{F}$ satisfies item (ii), and, by Proposition 48, does not change whether $\mathcal{F}$ satisfies item (i). Thus we may assume all signatures in $\mathcal{F}$ have even arity. For $F \in \mathcal{F}$, let $\widetilde{F}$ be the matrix of the binary signature constructed by contracting all but one pair of inputs of F (see Figure 13). Since

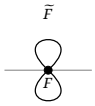


Figure 13: Illustrating (the gadget with signature) $\widetilde{F}$ for 6-ary F.

F is symmetric, $\widetilde{F}$ doesn't depend on how we pair up F's inputs. Every $\widetilde{F}$, and every composition $\widetilde{F_1} \circ \widetilde{F_2}$ for $F_1, F_2 \in \mathcal{F}$, is the signature of a connected $\mathcal{F}$-gadget, so is symmetric by assumption. Therefore, as in Remark 50, the matrices $\widetilde{F}$ for $F \in \mathcal{F}$ all commute.

Claim 51. *If* $\mathbf{K}$ *is a connected binary* $\mathcal{F}$*-gadget with* p *vertices, assigned signatures* $F_1, \ldots, F_p \in \mathcal{F}$*, then* $M(\mathbf{K}) = \prod_{i=1}^{p} \widetilde{F_i}$*.*

Proof. We prove Claim 51 by induction on p. For $p = 1$, by the symmetry of F_1, every connected binary F_1-gadget with a single vertex has signature $\widetilde{F_1}$. Now suppose $p \geq 2$. $\mathbf{K}$ contains a vertex v whose removal does not disconnect $\mathbf{K}$ (take e.g. the final vertex visited by breadth first search). Assume WLOG that v is assigned signature F_p. Construct $\mathbf{K'}$ from $\mathbf{K}$ by breaking all but one edge between v and other vertices (see Figure 14). Each broken edge becomes two new dangling edges. Since $\mathbf{K'}$ is connected, its signature is symmetric by assumption. The number of dangling edges incident to v is odd, as v has even degree and exactly one edge to another vertex (loops on v do not affect the parity). Since $\mathbf{K'}$ has an even number of dangling edges (two plus twice the number of edges broken), there are an odd number of dangling edges incident to the other vertices of $\mathbf{K'}$. Now create a binary gadget $\mathbf{K''}$ from $\mathbf{K'}$ by arbitrarily pairing up and connecting all but one dangling edge incident to v, and similarly pairing up and connecting all but one dangling edge incident to the other vertices of $\mathbf{K'}$. We may also recover $\mathbf{K}$ from $\mathbf{K'}$ by connecting possibly different pairs of dangling edges (reforming the edges broken to create $\mathbf{K'}$) and, since the signature of $\mathbf{K'}$ is symmetric, the signature of a gadget produced by connecting dangling edges

of $\mathbf{K}'$ does not depend on which pairs of dangling edges we connect (although the underlying graphs of the gadgets differ). Therefore $M(\mathbf{K}'') = M(\mathbf{K})$.

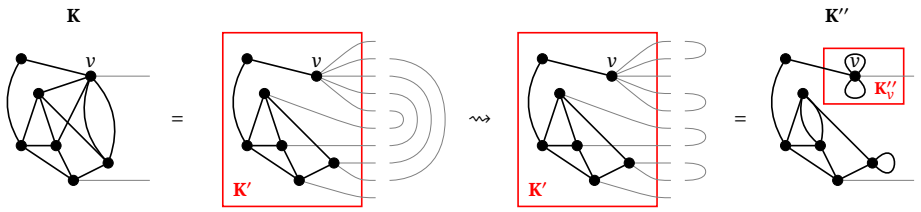


Figure 14: Illustrating the proof of Claim 51 when v has a single incident dangling edge in $\mathbf{K}$. The cases where v has zero or two incident dangling edges are similar.

Let $\mathbf{K}''_v$ be the subgadget of $\mathbf{K}''$ induced by v. $\mathbf{K}''_v$ has two dangling edges, one dangling in $\mathbf{K}''$, and one which, in $\mathbf{K}''$, connects v to a vertex in $\overline{\mathbf{K}''_v}$ (the complement of $\mathbf{K}''_v$), with the remaining edges incident to v paired into loops. Thus $M(\mathbf{K}''_v) = \widetilde{F_p}$. Similarly, $\overline{\mathbf{K}''_v}$ has two dangling edges, one dangling in $\mathbf{K}''$ and one incident to to v in $\mathbf{K}''$, so we recover $\mathbf{K}''$ by reconnecting the edge between $\mathbf{K}''_v$ and $\overline{\mathbf{K}''_v}$. Thus, applying induction to $\overline{\mathbf{K}''_v}$, which has $p-1$ vertices, we obtain

$$M(\mathbf{K}) = M(\mathbf{K}'') = M(\mathbf{K}''_v) \circ M(\overline{\mathbf{K}''_v}) = \widetilde{F_p} \circ \prod_{i=1}^{p-1} \widetilde{F}_i = \prod_{i=1}^{p} \widetilde{F}_i$$

(recall that the $\widetilde{F}_i$ commute). This completes the proof of Claim 51. □

The matrices $\widetilde{F}$ for $F \in \mathcal{F}$ are symmetric and commute, so they are simultaneously diagonalizable: there is an $H \in O(q)$ such that, for every $F \in \mathcal{F}$, $H\widetilde{F}H^\top = (=_2^{\mathbf{a}^F})$ for some $\mathbf{a}^F \in \mathbb{R}^q$. By Proposition 13 and part 2 of Proposition 34, we may replace $\mathcal{F}$ with $H\,\mathcal{F}$ to assume each $\widetilde{F} = (=_2^{\mathbf{a}^F})$. Define

$$\mathcal{G} = \{=^{\mathbf{a}^F}_{\text{arity}(F)} \mid F \in \mathcal{F}\} \subset \mathcal{GEQ},$$

a set similar to $\mathcal{F}$. Let Ω be a $\mathcal{F}$-grid, containing signatures $F_1, \ldots, F_p$. As Ω is not a tree (all of its vertices have even degree), we can break some edge of Ω to produce a connected binary $\mathcal{F}$-gadget $\mathbf{K}$. By Claim 51,

$$M(\mathbf{K}) = \prod_{i=1}^{p} \widetilde{F}_i = \prod_{i=1}^{p} (=_2^{\mathbf{a}^{F_i}}) = \left(=_2^{\mathbf{a}^{F_1} \bullet \ldots \bullet \mathbf{a}^{F_p}}\right) = M(\mathbf{K}_{\mathcal{F}\to\mathcal{G}}).$$

Now, reconnecting the dangling edges of $\mathbf{K}$, we find $\text{Holant}_\Omega = \text{Holant}_{\Omega_{\mathcal{F}\to\mathcal{G}}}$. Thus $\mathcal{F}$ and $\mathcal{G}$ are Holant-indistinguishable, so, by Theorem 12, $\mathcal{F}$ and $\mathcal{G}$ are ortho-equivalent. Hence $\mathcal{F}$ is odeco.

(iii) $\implies$ (ii): Let K be the n-ary signature of a connected $\mathcal{F}$-gadget $\mathbf{K}$. Every unary signature is trivially symmetric, so assume $n \geq 2$. It suffices to

show that, for any fixed partial input $\mathbf{z} \in [q]^{n-2}$ to K and any $x, y \in [q]$, we have $K(x, y, \mathbf{z}) = K(y, x, \mathbf{z})$ (where we assume WLOG that x and y are the first two inputs to K by reordering the dangling edges of $\mathbf{K}$). Let u and w be the vertices of $\mathbf{K}$ incident to the first and second dangling edges of $\mathbf{K}$ (after reordering). If $u = w$ then we are done, as every $F \in \mathcal{F}$ is symmetric. Otherwise, since $\mathbf{K}$ is connected, it contains a path $P = (u = v_0, v_1, \ldots, v_{p-2}, v_{p-1} = w)$ from u to w, where v_i is assigned signature $F_i \in \mathcal{F}$, for $i \in [p]$. Let $E(P) := \{e_0, e_1, \ldots, e_{p-1}, e_p\}$ be the edges of P, including the dangling edges e_0 and e_p incident to u and w, respectively. Then e_i and e_{i+1} are inputs to F_i for all $i \in [p]$. For any fixed assignment $\sigma : E(\mathbf{K}) \setminus E(P) \to [q]$, define the matrix $F_i^\sigma \in \mathbb{R}^{q \times q}$ by $(F_i^\sigma)_{a,b} := F_i(\sigma|_{\delta(v_i)}, a, b)$ (that is, fix the inputs to F_i from edges outside of $E(P)$). Then

$$K(x, y, \mathbf{z}) = \sum_{\substack{\sigma : E(\mathbf{K}) \setminus E(P) \to [q] \\ \sigma(D) = \mathbf{z}}} \left(\prod_{v \in V(\mathbf{K}) \setminus P} F_v(\sigma|_{\delta(v)}) \right) F_P^\sigma(x, y), \tag{18}$$

where D is the ordered list of the last $n-2$ dangling edges of $\mathbf{K}$ and

$$\begin{aligned} F_P^\sigma(x, y) &= \sum_{\substack{\phi : E(P) \to [q] \\ \phi(e_0) = x, \phi(e_p) = y}} \prod_{i=0}^{p-1} F_i(\sigma|_{\delta(v_i)}, \phi(e_i), \phi(e_{i+1})) \\ &= \sum_{\substack{\phi : E(P) \to [q] \\ \phi(e_0) = x, \phi(e_p) = y}} \prod_{i=0}^{p-1} (F_i^\sigma)_{\phi(e_i), \phi(e_{i+1})} = \left(\prod_{i=0}^{p-1} F_i^\sigma \right)_{x,y}. \end{aligned}$$

On the RHS of (18), x and y appear only in $F_P^\sigma(x, y)$. Thus it suffices to show that, for any fixed σ, $F_P^\sigma(x, y) = F_P^\sigma(y, x)$. For any $i, j \in [p]$ and $a, b \in [q]$,

$$\begin{aligned} &(F_i^\sigma F_j^\sigma)_{a,b} = \sum_{z \in [q]} (F_i^\sigma)_{a,z} (F_j^\sigma)_{z,b} = \sum_{z \in [q]} F_i(\sigma|_{\delta(v_i)}, a, z) F_j(\sigma|_{\delta(v_j)}, b, z) = (F_i * F_j)(\sigma|_{\delta(v_i)}, a, \sigma|_{\delta(v_j)}, b) \\ &= (F_i * F_j)(\sigma|_{\delta(v_i)}, b, \sigma|_{\delta(v_j)}, a) = \sum_{z \in [q]} F_i(\sigma|_{\delta(v_i)}, b, z) F_j(\sigma|_{\delta(v_j)}, a, z) = \sum_{z \in [q]} (F_i^\sigma)_{b,z} (F_j^\sigma)_{z,a} = (F_i^\sigma F_j^\sigma)_{b,a}, \end{aligned}$$

where the fourth equality uses the assumption that $F_i * F_j$ is symmetric. Thus $F_i^\sigma F_j^\sigma$ is symmetric. Both F_i^σ and F_j^σ are symmetric, as F_i and F_j are symmetric, so, as in Remark 50, F_i^σ and F_j^σ commute. Therefore

$$F_P^\sigma(x, y) = \left(\prod_{i=0}^{p-1} F_i^\sigma \right)_{x,y} = \left(\left(\prod_{i=0}^{p-1} F_i^\sigma \right)^\top \right)_{y,x} = \left(\prod_{i=0}^{p-1} (F_{p-1-i}^\sigma)^\top \right)_{y,x} = \left(\prod_{i=0}^{p-1} F_i^\sigma \right)_{y,x} = F_P^\sigma(y, x).$$

□

6 Possible Variations of Theorem 12

6.1 Complex-valued signatures

Recall from Remark 15 that Theorem 12 does not hold for complex-valued $\mathcal{F}$ and $\mathcal{G}$, as observed by Draisma and Regts [24]. Draisma and Regts also note that a direct extension of Schrijver's duality (Theorem 29) to the complex orthogonal group $O_q(\mathbb{C})$ is impossible because $O_q(\mathbb{C})$ is not compact. However, they also provide some techniques for handling complex-valued signatures in edge coloring models, including a version [24, Theorem 3] of Theorem 37 for edge-coloring models parameterized by complex-valued sets $\mathcal{F}$ that are *nondegenerate*, meaning that the natural bilinear form on the complex tensor algebra $T(\mathbb{C}^q)$ is nondegenerate on the subalgebra $[\mathcal{F}]$ of quantum gadget signatures.

Cai and Young [10, 55] prove their #CSP indistinguishability theorems for complex-valued signature sets with the assumption that the sets are *conjugate-closed*, meaning they must contain the entrywise conjugate of each of their complex signatures (the counterexample set $\mathcal{F} = \{[1, i]\}$ from Remark 15 is not conjugate-closed). In particular, conjugate-closed $\mathcal{F}$ are nondegenerate. After the publication of the first version [54] of this work, Cai and Young [11] showed that Theorem 12 indeed holds for complex orthogonal H and nondegenerate (or *quantum-nonvanishing*) $\mathcal{F}$ and $\mathcal{G}$. They achieved this generalization by applying an extension, due to Derksen and Makam [23], of Theorem 29 to nondegenerate complex-valued sets, leading to a similar extension of Theorem 37.

6.2 Quantum orthogonal matrices and planar signature grids

A *quantum orthogonal matrix* U is, roughly, a matrix whose entries come from an abstract, not necessarily commutative, C^*-algebra, satisfying the relation $UU^\top = U^\top U = I \otimes \mathbf{1}$, where $\mathbf{1}$ is the identity element of the C^*-algebra. Just as a permutation matrix is an orthogonal matrix that stabilizes $\mathcal{EQ}$, a *quantum permutation matrix* is a quantum orthogonal matrix that stabilizes $\mathcal{EQ}$ (see e.g. [10, Equation 27]). The main theorem of Cai and Young [10], extending the result of Mančinska and Roberson [36], is a planar, quantum version of Corollary 40: $\mathcal{F}$ and $\mathcal{G}$ are Pl-Holant$(\cdot \cup \mathcal{EQ})$-indistinguishable (planar-#CSP-indistinguishable) if and only if there is a quantum permutation matrix U satisfying $U\,\mathcal{F} = \mathcal{G}$ ($\mathcal{F}$ and $\mathcal{G}$ are *quantum isomorphic*). Removing $\mathcal{EQ}$, we should obtain the following planar, quantum version of Theorem 12.

Conjecture 52. *Let $\mathcal{F}$, $\mathcal{G}$ be sets of real-valued signatures. Then the following are equivalent.*

(i) Pl-Holant$_\Omega(\mathcal{F}) =$ Pl-Holant$_{\Omega_{\mathcal{F}\to\mathcal{G}}}(\mathcal{G})$ *for every planar $\mathcal{F}$-grid Ω.*

(ii) There is a quantum orthogonal matrix U such that $U\,\mathcal{F} = \mathcal{G}$.

Cai and Young [10, Theorem 5] prove (ii) $\implies$ (i) when U is a quantum permutation matrix; however, their proof only relies on U being a quantum

orthogonal matrix. Therefore, to prove Conjecture 52, it suffices to show (i) $\implies$ (ii). The *Tannaka-Krein duality* for the quantum symmetric group used by Mančinska and Roberson and Cai and Young [10, Theorem 3] has a more general version for the quantum orthogonal group O_q^+ [36, Theorem 2.13]. This version is analogous to Theorem 29 above, but concerning quantum subgroups of O_q^+ and asymmetric (planar) tensor categories with duals. Then, following the proof of [10, Theorem 4], but ommitting the gadgets $E^{1,0}$ and $E^{1,2}$ used to construct $\mathcal{EQ}$, we will obtain a quantum analogue of our Theorem 37 for planar quantum $\mathcal{F}$-gadgets, giving a quantum analogue of Lemma 39. The rest of the proof of Theorem 12, however, involves nonplanar signature grid manipulations (e.g. in Figure 5 it is in general impossible to embed Ω such that every instance of D lies on the outer face) and, more critically, relies on the existence of the singular value decomposition of a submatrix of a real orthogonal matrix, then on viewing the resulting diagonal matrix as a signature. It is yet unclear whether the same or similar reasoning applies to a submatrix of a quantum orthogonal matrix. However, after publication of the first version [54] of this work, Conjecture 52 was confirmed using different techniques in the author's Ph.D. thesis [56].

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A Block Signature Actions

In this appendix, we prove several technical results which state that the action of a block matrix H on a block signature K follows block matrix multiplication rules as one would expect.

Proposition 53. *Let $\mathcal{I} = X \sqcup Y$. For $H \in \mathbb{R}^{\mathcal{I} \times \mathcal{I}}$, $K^{m,d} \in \mathbb{R}^{\mathcal{I}^m \times \mathcal{I}^d}$ and any $\mathbf{R} \in \{X,Y\}^m$ $\mathbf{C} \in \{X,Y\}^d$,*

$$(H^{\otimes m} K^{m,d})|_{\mathbf{R},\mathbf{C}} = \sum_{\mathbf{J} \in \{X,Y\}^m} \left(\bigotimes_{i=1}^{m} H|_{R_i,J_i} \right) K^{m,d}|_{\mathbf{J},\mathbf{C}}$$

(with $H^{\otimes m} K^{m,d} \in \mathbb{R}^{\mathcal{I}^m \times \mathcal{I}^d}$ indexed as in part 3 of Definition 16) and similarly

$$(K^{m,d} H^{\otimes d})|_{\mathbf{R},\mathbf{C}} = \sum_{\mathbf{J} \in \{X,Y\}^d} K^{m,d}|_{\mathbf{R},\mathbf{J}} \left(\bigotimes_{i=1}^{d} H|_{J_i,C_i} \right).$$

That is, with $H^{\otimes m}|_{\mathbf{R},\mathbf{J}} = \bigotimes_{i=1}^{m} H|_{R_i,J_i}$, we can compute $H^{\otimes m} K^{m,d}$ as a block matrix product with

$$H^{\otimes m} = \begin{bmatrix} H^{\otimes m}|_{X^m,X^m} & H^{\otimes m}|_{X^m,X^{m-1}Y} & H^{\otimes m}|_{X^m,X^{m-2}YX} & \cdots & H^{\otimes m}|_{X^m,Y^m} \\ H^{\otimes m}|_{X^{m-1}Y,X^m} & H^{\otimes m}|_{X^{m-1}Y,X^{m-1}Y} & H^{\otimes m}|_{X^{m-1}Y,X^{m-2}YX} & \cdots & H^{\otimes m}|_{X^{m-1}Y,Y^m} \\ \vdots & \vdots & \vdots & & \vdots \\ H^{\otimes m}|_{Y^{m-1}X,X^m} & H^{\otimes m}|_{Y^{m-1}X,X^{m-1}Y} & H^{\otimes m}|_{Y^{m-1}X,X^{m-2}YX} & \cdots & H^{\otimes m}|_{Y^{m-1}X,Y^m} \\ H^{\otimes m}|_{Y^m,X^m} & H^{\otimes m}|_{Y^m,X^{m-1}Y} & H^{\otimes m}|_{Y^m,X^{m-2}YX} & \cdots & H^{\otimes m}|_{Y^m,Y^m} \end{bmatrix}$$

and

$$K^{m,d} = \begin{bmatrix} K^{m,d}|_{X^m,X^d} & K^{m,d}|_{X^m,X^{d-1}Y} & \cdots & K^{m,d}|_{X^m,Y^{d-1}X} & K^{m,d}|_{X^m,Y^d} \\ K^{m,d}|_{X^{m-1}Y,X^d} & K^{m,d}|_{X^{m-1}Y,X^{d-1}Y} & \cdots & K^{m,d}|_{X^{m-1}Y,Y^{d-1}X} & K^{m,d}|_{X^{m-1}Y,Y^d} \\ K^{m,d}|_{X^{m-2}YX,X^d} & K^{m,d}|_{X^{m-2}YX,X^{d-1}Y} & \cdots & K^{m,d}|_{X^{m-2}YX,Y^{d-1}X} & K^{m,d}|_{X^{m-2}YX,Y^d} \\ \vdots & \vdots & & \vdots & \vdots \\ K^{m,d}|_{Y^m,X^d} & K^{m,d}|_{Y^m,X^{d-1}Y} & \cdots & K^{m,d}|_{Y^m,Y^{d-1}X} & K^{m,d}|_{Y^m,Y^d}. \end{bmatrix}$$

Proof. We prove the first statement. The second is proved similarly. Let $\mathbf{r} \in \mathbf{R}$

and $\mathbf{c} \in \mathbf{C}$. Then

$$\begin{aligned}((H^{\otimes m}K^{m,d})|_{\mathbf{R},\mathbf{C}})_{\mathbf{r},\mathbf{c}} = (H^{\otimes m}K^{m,d})_{\mathbf{r},\mathbf{c}} &= \sum_{\mathbf{j}\in\mathcal{I}^m}\left(\prod_{i=1}^m H_{r_i,j_i}\right)K^{m,d}_{\mathbf{j},\mathbf{c}} \\ &= \sum_{\mathbf{J}\in\{X,Y\}^m}\sum_{\mathbf{j}\in\mathbf{J}}\left(\prod_{i=1}^m H_{r_i,j_i}\right)K^{m,d}_{\mathbf{j},\mathbf{c}} \\ &= \sum_{\mathbf{J}\in\{X,Y\}^m}\sum_{\mathbf{j}\in\mathbf{J}}\left(\bigotimes_{i=1}^m H|_{R_i,J_i}\right)_{\mathbf{r},\mathbf{j}}(K^{m,d}|_{\mathbf{J},\mathbf{C}})_{\mathbf{j},\mathbf{c}} \\ &= \sum_{\mathbf{J}\in\{X,Y\}^m}\left(\left(\bigotimes_{i=1}^m H|_{R_i,J_i}\right)K^{m,d}|_{\mathbf{J},\mathbf{C}}\right)_{\mathbf{r},\mathbf{c}} \\ &= \left(\sum_{\mathbf{J}\in\{X,Y\}^m}\left(\bigotimes_{i=1}^m H|_{R_i,J_i}\right)K^{m,d}|_{\mathbf{J},\mathbf{C}}\right)_{\mathbf{r},\mathbf{c}}\square\ .\end{aligned}$$

We will only need Proposition 53 as written, for partitions of $\mathcal{I}$ into two blocks, but it is not hard to see that it extends naturally to partitions of $\mathcal{I}$ into more than two blocks.

We will apply Proposition 53 for two special types of H.

Corollary 54. *If*

$$H = H_X \oplus H_Y = \begin{bmatrix} H_X & 0 \\ 0 & H_Y \end{bmatrix}$$

is block-diagonal, then, for any $K \in \mathbb{R}^{\mathcal{I}^n}$, the block form of $H^{\otimes n}K^{n,0}$ is

$$\begin{bmatrix} H_X^{\otimes n} & 0 & \dots & 0 & 0 \\ 0 & * & \dots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \dots & * & 0 \\ 0 & 0 & \dots & 0 & H_Y^{\otimes n} \end{bmatrix}\begin{bmatrix}(K|_X)^{n,0} \\ * \\ \vdots \\ * \\ (K|_Y)^{n,0}\end{bmatrix} = \begin{bmatrix} H_X^{\otimes n}(K|_X)^{n,0} \\ * \\ \vdots \\ * \\ H_Y^{\otimes n}(K|_Y)^{n,0}\end{bmatrix}.$$

Proof. Here, $H|_{R_i,J_i} = H_{R_i}$ if $R_i = J_i$ and $H|_{R_i,J_i} = 0$ if $R_i \neq J_i$, so H takes the claimed block form in Proposition 53. $\square$

If $X = V(F)$, $Y = V(G)$, and $K = F \oplus G$, then by (2), all blocks of K are 0 except $K|_X = F$ and $K|_Y = G$, so, by Corollary 54,

$$(H_{V(F)} \oplus H_{V(G)})(F \oplus G) = (H_{V(F)}F) \oplus (H_{V(G)}G). \tag{19}$$

Corollary 55. *If*

$$B = \begin{bmatrix} 0 & H^\top \\ H & 0 \end{bmatrix}$$

is block-antidiagonal, with blocks indexed by $X = V(F)$ and $Y = V(G)$, then, for any m, d,

$$H^{\otimes m}F^{m,d} = G^{m,d}H^{\otimes d} \ \text{ and } \ (H^\top)^{\otimes m}G^{m,d} = F^{m,d}(H^\top)^{\otimes d} \iff B^{\otimes m}(F\oplus G)^{m,d} = (F\oplus G)^{m,d}B^{\otimes d}.$$

Proof. $B|_{R_i,J_i} = 0$ if $R_i = J_i$, so, by Proposition 53 and (2), we compute blockwise with

$$B^{\otimes m}(F \oplus G)^{m,d} \text{ as } \begin{bmatrix} 0 & 0 & \dots & 0 & (H^\top)^{\otimes m} \\ 0 & 0 & \dots & * & 0 \\ \vdots & \vdots & \iddots & \vdots & \vdots \\ 0 & * & \dots & 0 & 0 \\ H^{\otimes m} & 0 & \dots & 0 & 0 \end{bmatrix} \begin{bmatrix} F^{m,d} & 0 & \dots & 0 & 0 \\ 0 & 0 & \dots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \dots & 0 & 0 \\ 0 & 0 & \dots & 0 & G^{m,d} \end{bmatrix} \tag{20}$$

and

$$(F \oplus G)^{m,d} B^{\otimes d} \text{ as } \begin{bmatrix} F^{m,d} & 0 & \dots & 0 & 0 \\ 0 & 0 & \dots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \dots & 0 & 0 \\ 0 & 0 & \dots & 0 & G^{m,d} \end{bmatrix} \begin{bmatrix} 0 & 0 & \dots & 0 & (H^\top)^{\otimes d} \\ 0 & 0 & \dots & * & 0 \\ \vdots & \vdots & \iddots & \vdots & \vdots \\ 0 & * & \dots & 0 & 0 \\ H^{\otimes d} & 0 & \dots & 0 & 0 \end{bmatrix}. \tag{21}$$

The result follows from comparing the bottom left and upper right blocks of (20) and (21). □